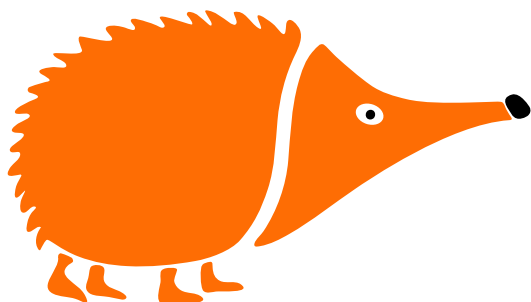




EchidnaBlack

Documentación archivada de echidna.es

Echidna Educación · echidna.es · octubre de 2026

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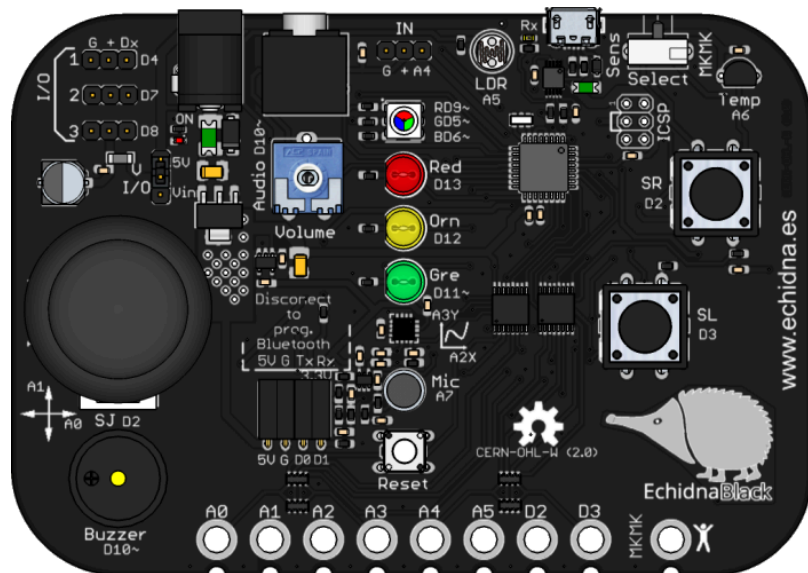
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La placa

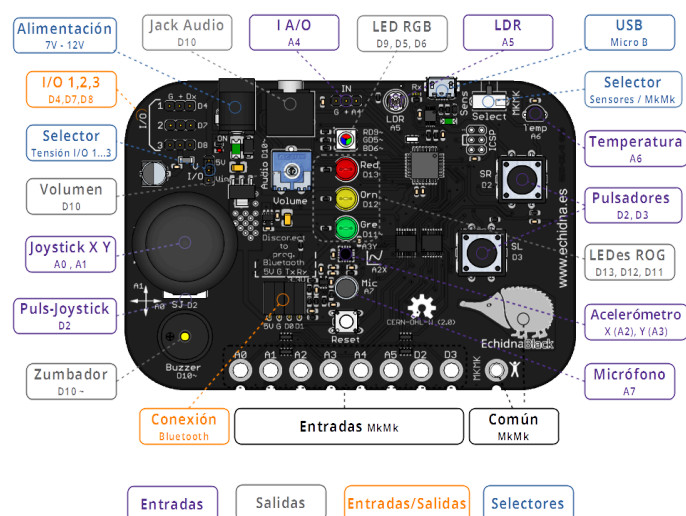
Presentación

Características



- Placa autónoma compatible con Arduino Nano/ Arduino UNO
- Sensores integrados: pulsadores, joystick, acelerómetro, luz, temperatura, micrófono
- Actuadores integrados: LEDs, LED RGB, audio
- Flexible: entradas y salidas disponibles para complementos
- 8 Entradas tipo Makey Makey
- Conexión BlueTooth
- Open Source Hardware Certification con registro OSHWA UID ES000010

Esquema de componentes



Componentes

Pulsadores

Es un componente electromecánico que permite abrir o cerrar un circuito con un solo estado estable

Joystick

Es un mando que consta de dos potenciómetros uno para el eje X y otro para el eje Y.

Sensor Luz LDR

“Light Dependent Resistor”, es una resistencia cuyo valor depende de la cantidad de luz que incide sobre ella.

LED RGB

Light Emitting Diode y Red Green Blue. Son 3 LEDes Rojo, Verde y Azul en la misma cápsula.

Audio

Dispone de dos salidas para reproducir audio: el zumbador y el jack, ambos con un regulador de volumen.

MkMk

Echidna dispone de 8 Conectores MkMk que le permiten detectar gran variedad de objetos.

Acelerómetro

Sensor de aceleraciones basado en condensadores dentro de una estructura

LEDs ROG

Light Emitting Diode se usan como testigos (indicadores) y como fuente de iluminación.

Micrófono

Transductor que nos entrega una señal eléctrica de similares características al sonido.

Sensor Temperatura

Es un sensor de temperatura calibrado cuya salida es lineal, y donde cada 10 mV equivale a 1°C.

Puesta en marcha

Modo Sensores/Modo MkMk

Alimentación

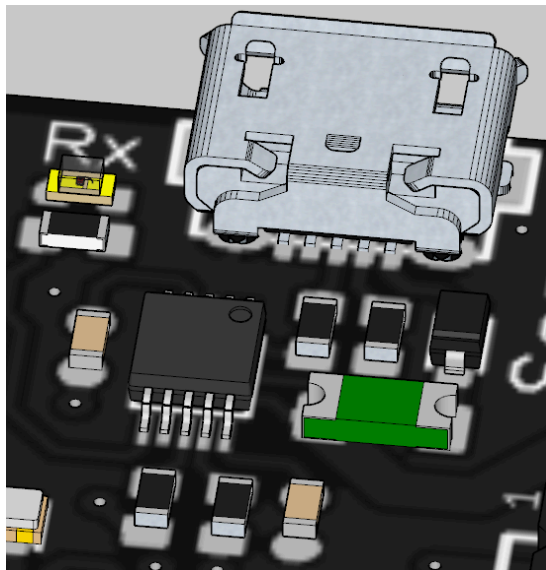
Complementos

Documentación

Puesta en marcha

Mediante USB podemos conectar nuestra Echidna Black al ordenador, dándole alimentación y permitiendo que el PC se comunique con ella. La comunicación con el PC de la Echidna Black se realiza mediante el programa [StandardFirmata](#) ya instalado y con el [Bootloader Optiboot](#) configurado como Arduino Nano que nos permite tener acceso a “A6 y A7”.

Para que Echidna Black se pueda comunicar con nuestro PC es necesario que nuestro sistema operativo (SO) le dé permiso de acceso al puerto serie (USB) y que tenga el driver del controlador de comunicación (CH340) instalado.



- Driver de comunicación:

Echidna Black utiliza el chip CH340E, por lo que dependiendo del SO necesitarás instalar el controlador “[Driver CH341](#)”

- GNU Linux: En caso de que seas usuario Linux, no debería ser necesario instalar el driver. Si necesitas el [driver para GNU Linux](#).
- MAC: Aquí tienes acceso al [driver para MAC](#).
- Windows: Aquí tienes acceso al [driver para Windows](#).
- Android: Aquí tienes acceso al [driver para Android](#).

- Permiso de acceso al puerto serie:

Dependiendo del SO es necesario dar o no permisos de acceso al puerto serie. Si ya has instalado el IDE de Arduino, estos permisos deberían estar ya dados.

- GNU Linux: Si no no tuvieras acceso al puerto serie puede que tengas que darle permiso desde una terminal usando el comando: `sudo usermod -a -G dialout «usuario»`.
- MAC: por defecto ya tenemos acceso al puerto serie.
- Windows: por defecto ya tenemos acceso al puerto serie.

Ahora ya puedes conectar EchidnaBlack a tu software favorito. Si quieres también dispones de las instrucciones para [instalar el Standardfirmata](#).

Si quisieras modificar el programa que viene instalado en Echidna Black, podrás hacerlo con Arduino. A continuación te contamos cómo instalar Arduino:

– **Instalar Arduino:**

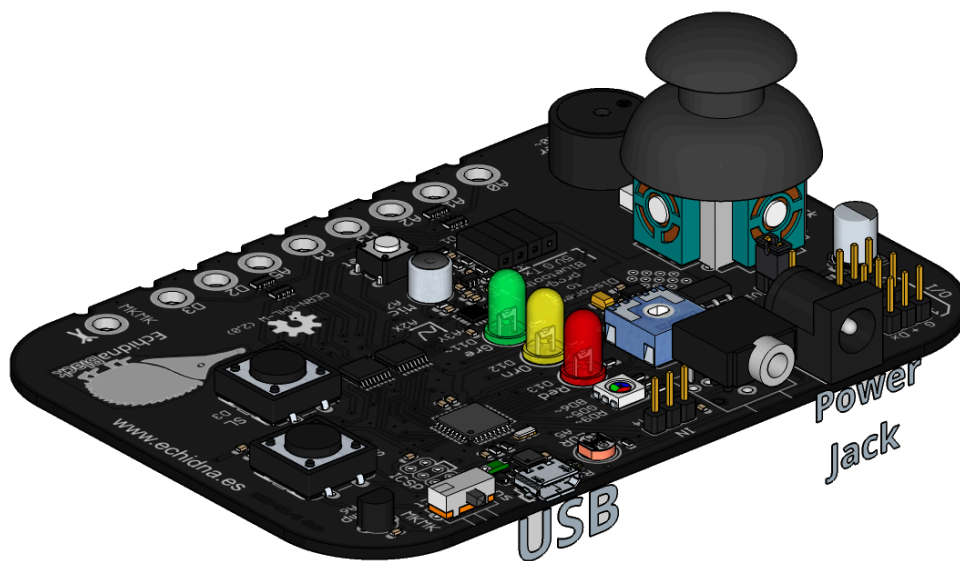
Para instalar Arduino nos debemos [descargar el IDE](#) de la página oficial:

En la guía [Getting Started](#) tenemos indicaciones sobre como instalar Arduino según nuestro sistema operativo.

Alimentación

Formas de alimentar la placa

1. Mediante la **conexión USB**: esta forma es válida para la mayoría de las funciones, toda la placa recibe una tensión regulada de 5V proporcionada por el cable USB, generalmente con un límite de 500mA, y cuenta con fusible rearmable.
2. Mediante el **Jack de alimentación**: Todas las partes reciben una tensión regulada internamente de 5V y 1A máx. Nos da la posibilidad de seleccionar Vin para alimentar los pines I/O con la potencia que proporciona el alimentador externo cuenta con fusible rearmable de protección (no superar 1A de corriente constante).



Selector de alimentación

El jumper de alimentación permite seleccionar la alimentación de las I/O.





Selector 5v

En caso de querer alimentar las I/O desde los 5 Volts procedentes del Arduino el jumper debe estar colocado en esta posición. Aconsejado para sensores externos que necesiten una tensión estabilizada.

¡No utilizar la alimentación 5V cuando los servos a controlar consuman más de 300 mA!* De lo contrario se sobrepasaría el regulador.

Selector alimentación Vin (Alimentación externa)

Aconsejado para alimentar servos u otros dispositivos conectados a I/O que consuman más de 300 mA con alimentación externa por el Jack de alimentación.

La alimentación de los servos cuenta con un filtro L-C para evitar que lleguen los parásitos de los motores al procesador.

Modo sensores / Modo MkMk

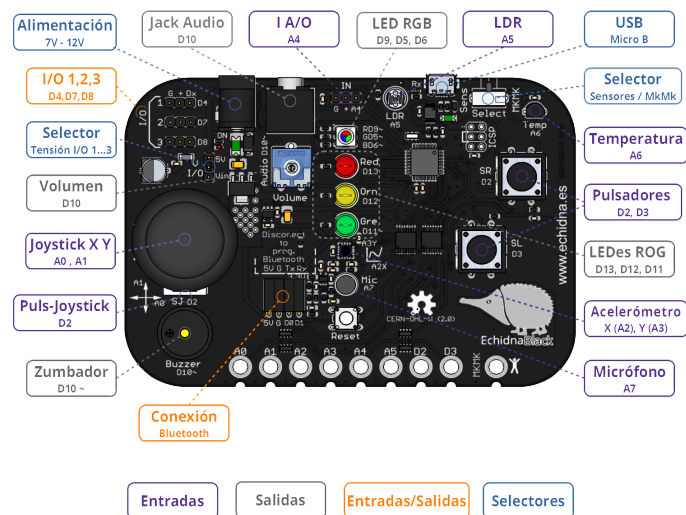
Selector Modo Sensores/ Modo MkMk



Es un conmutador que selecciona entre el Modo Sensores y el Modo Makey Makey.

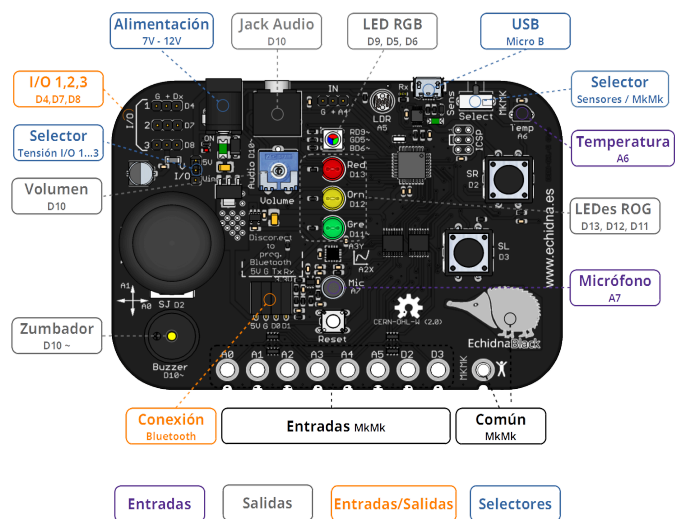
Modo sensores

En el modo sensores tenemos activo todos los componentes de la placa exceptuando las conexiones MkMk.



Modo MkMk

En este modo tenemos activas todas las salidas, las I/O y las conexiones MkMk.

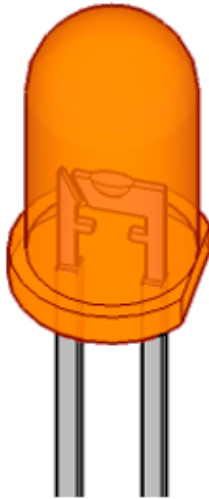


Componentes

LEDs

Descripción

Acrónimo de **Light Emitting Diode**, basado en el fenómeno de electroluminiscencia. Se usan como testigos (indicadores) y como fuente de iluminación.



Funcionamiento

Al aplicar tensión emite luz, por lo tanto podemos controlar su encendido apagado en digital “1” enciende, “0” apaga y su brillo en las salidas PWM

Todos los LEDes se conectan con una resistencia serie “ R_s ” para controlar la tensión/corriente aplicada.



Pines

- D11~ «Gre»: salida digital y PWM
- D12 «Orn»: salida digital

- D13 «Red»: salida digital

Digital: se enciende con un «1» y se apaga con un «0».

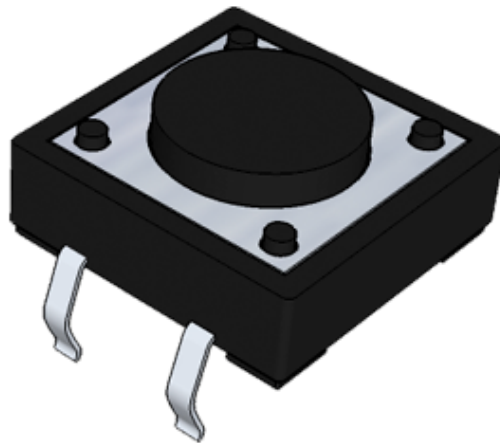
PWM: en la salida D11~ «Gre» se puede regular la intensidad luminosa con un valor de 0 a 255.

Hoja de características (anexo 1)

Pulsadores

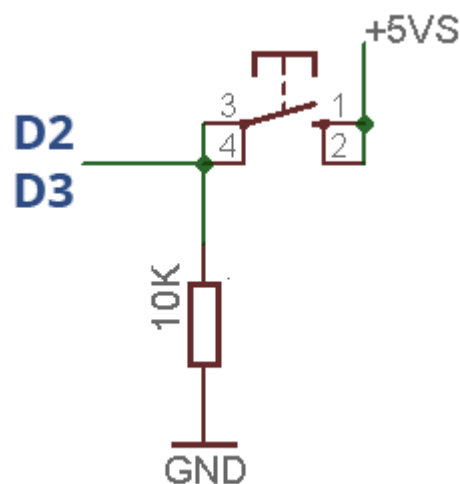
Descripción

Es un componente electromecánico que permite abrir o cerrar un circuito con un solo estado estable.



Funcionamiento

Está conectado con una resistencia a masa formando un circuito denominado pull-down, que proporciona un “0” (0V) sin pulsar y un “1” (5V) cuando pulsamos.



Pines

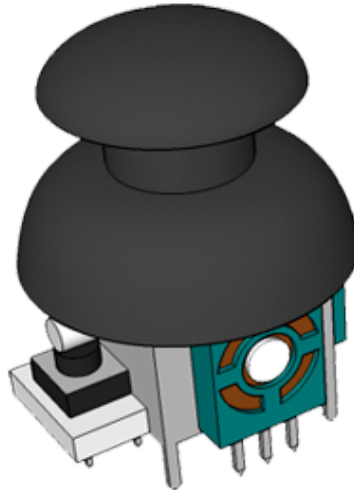
Entradas digitales: **D2** «SR» y **D3** «SL». Dan 0 si el pulsador no está pulsado y 1 si está pulsado.

Hoja de características (anexo 2)

Joystick

Descripción

Es un mando que consta de dos potenciómetros uno para el eje X y otro para el eje Y. Este modelo cuenta además con un pulsador.

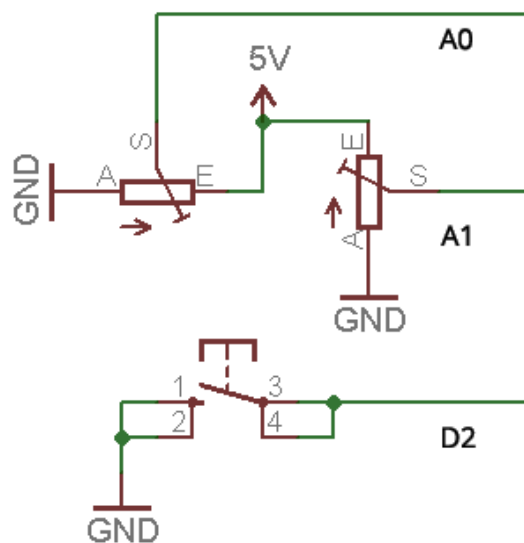


Funcionamiento

Permite transferir el movimiento del mando en una tensión proporcional entre 0-4,88V (0-999 de lectura analógica) * en la salida de cada eje (X, Y).

En la posición de reposo los valores rondan 2,5V (520 de lectura analógica) *, el pulsador del joystick está conectado con el pulsador «SR» (D2).

* Debido a las tolerancias de los componentes estos valores pueden ser distintos en cada placa.



Pines

- A0: eje X del joystick, entrada analógica
- A1: eje Y del joystick, entrada analógica
- D2: pulsador del joystick, entrada digital

Valores que da el joystick:

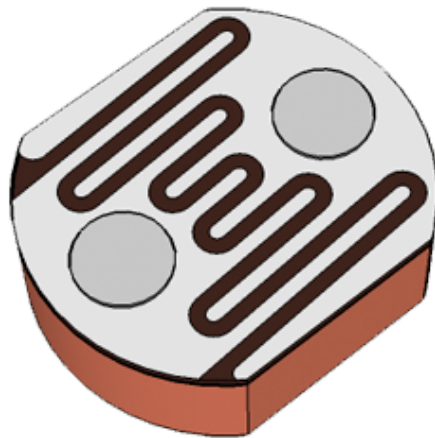
- En reposo, valores en torno a 512 en los dos ejes.
- En el eje X, 0 a la izquierda y 1023 a la derecha.
- En el eje Y, 0 abajo y 1023 arriba.

Hoja de características (anexo 3)

Sensor de luz (LDR)

Descripción

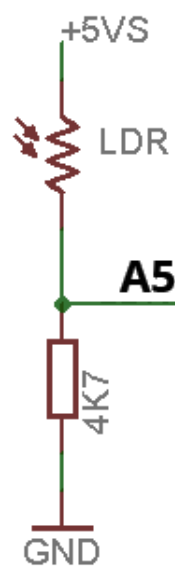
LDR es el acrónimo de “Light Dependent Resistor”, es una resistencia cuyo valor depende de la cantidad de luz que incide sobre ella. Generalmente se fabrica con sulfuro de cadmio.



Funcionamiento

La LDR proporciona valores altos de resistencia (en torno al $M\Omega$) con poca luz y valores bajos con mucha luz. Gracias a conectarla con una resistencia en serie forma un divisor de tensión que consigue invertir la lógica del funcionamiento, de forma que con mucha luz proporciona valores altos de tensión $4,9V = 999$ (de lectura analógica) * y valores bajos con poca luz ($0,0V = 000$).

* Debido a las tolerancias de los componentes estos valores pueden ser distintos en cada placa.



Pines

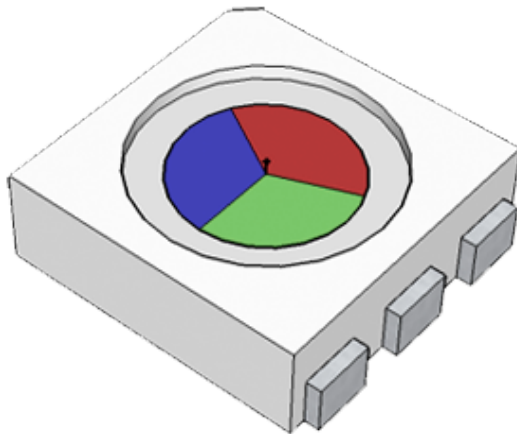
A5 «LDR»: entrada analógica.

Hoja de características

LED RGB

Descripción

LED RGB acrónimos de Light Emitting Diode y Red Green Blue, es decir son tres LEDes Rojo, Verde y Azul en la misma cápsula.



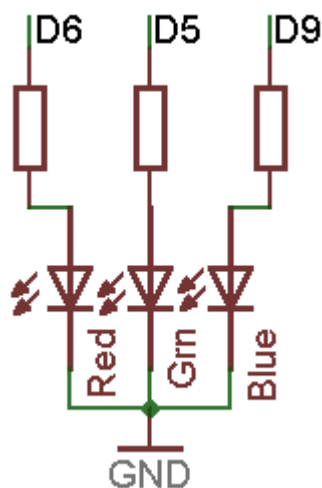
Funcionamiento

El LED RGB podemos controlarlo digitalmente, encendiendo o apagando cada LED y ajustar la luminosidad mediante PWM “analógico”.

Control digital: podemos encender y apagar cada uno de los LED RGB, lo que nos permite formar $2^3=8$ colores diferentes.

Control “analógico” PWM: podemos controlar el brillo de cada LED, variando su intensidad luminosa desde 0 a 255, lo que permite formar $256 \times 256 \times 256$, más de 16 millones de colores.

Todos los LEDes se conectan con una resistencia serie “Rs” para controlar la tensión/corriente aplicada.



Pines

- D9~ «RGB_R»: salida digital y PWM
- D5~ «RGB_G»: salida digital y PWM
- D6~ «RGB_B»: salida digital y PWM

Como salidas PWM, con un valor entre 0 y 255 se regula la intensidad de cada LED y, con ella, el color que emite.

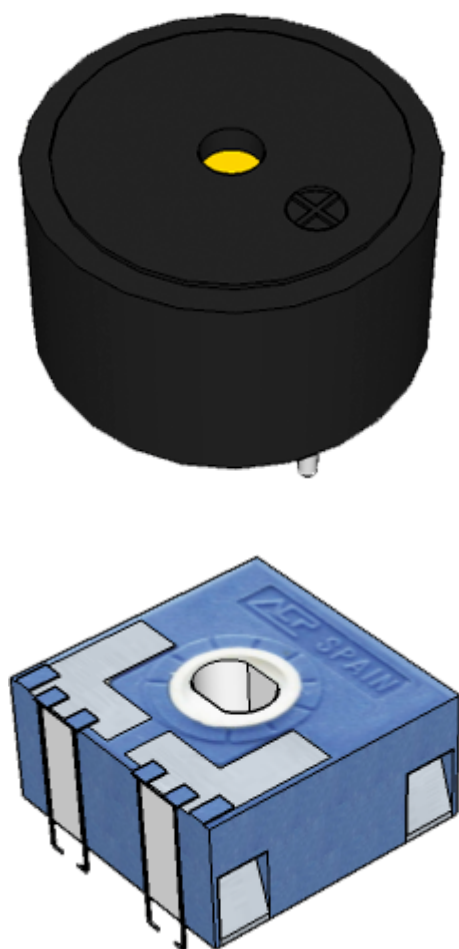
Hoja de características (anexo 4)

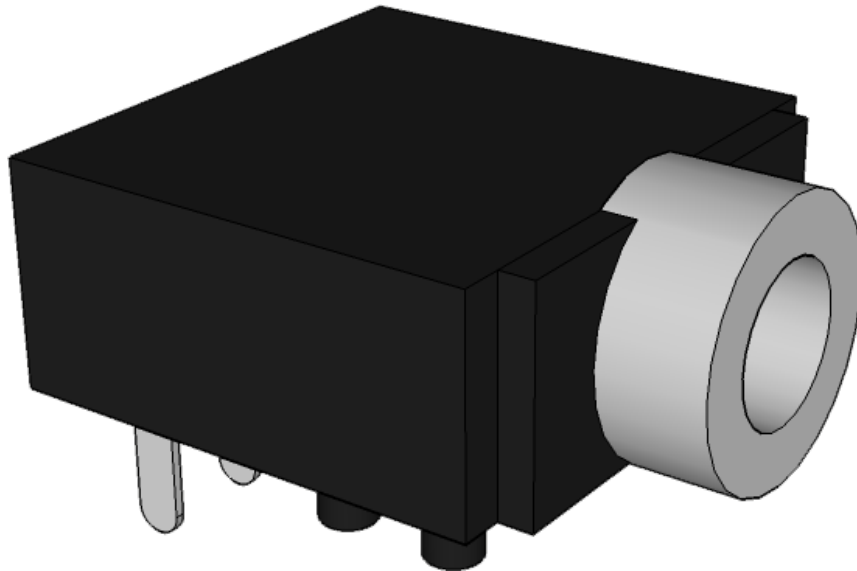
Audio

Descripción

Disponemos de dos salidas para reproducir audio, el zumbador y el jack al que podemos conectar auriculares o altavoces autoamplificados. Al conectar una clavija de audio en el jack se desconecta el zumbador.

Además contamos con un potenciómetro que permite ajustar el volumen del sonido.



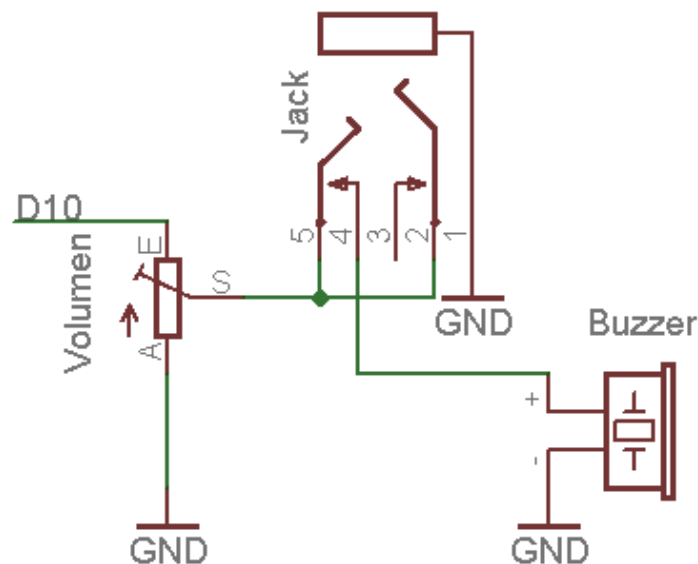


Funcionamiento

En D10~ podemos reproducir señales entre 31 Hz y 20 KHz.

El zumbador al ser excitado por una señal con una frecuencia determinada vibra reproduciendo un sonido. Hay que tener en cuenta que su frecuencia central es de 2,3 KHz, con lo que si nos alejamos mucho de esa frecuencia no sonará con calidad. (Do = C4 = 262 Hz) (cuidado con el volumen, a bajo nivel el zumbador puede no resultar audible).

En la salida del jack conectamos un equipo externo de audio que nos permite reproducir las frecuencias de entrada con más calidad (cuidado con el volumen, puede dañar los auriculares e incluso llegar al umbral doloroso en los oídos).



Pines

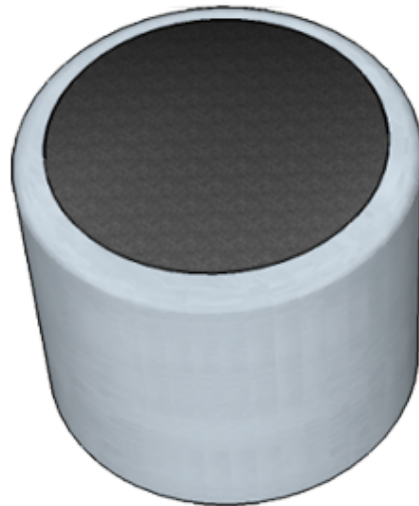
D10~ «Buzz»: salida PWM. Con un valor entre 0 y 255 se modula la frecuencia del sonido.

Hoja de características (anexo 5)

Micrófono

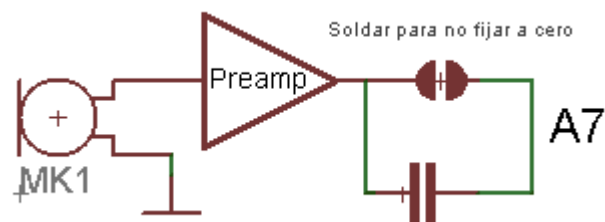
Descripción

Transductor acústico-eléctrico piezoeléctrico que nos entrega una señal eléctrica de similares características que el sonido que recibe.

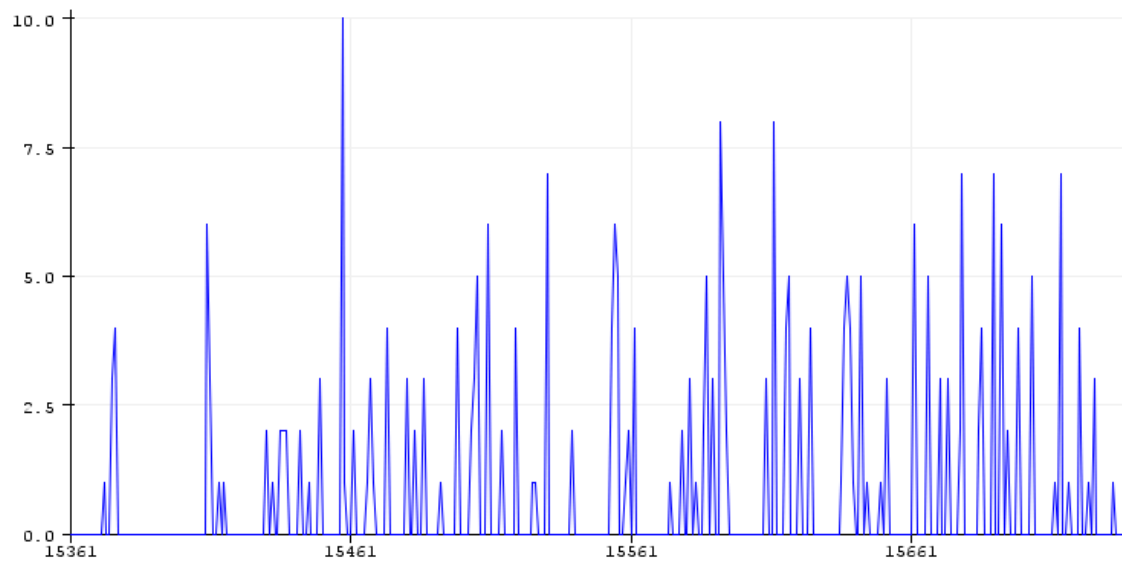


Funcionamiento

El micrófono entrega una señal que es necesario amplificar mediante circuitos electrónicos y se conecta a la entrada analógica A7, dispone de un puente de soldadura que permite fijar el nivel.



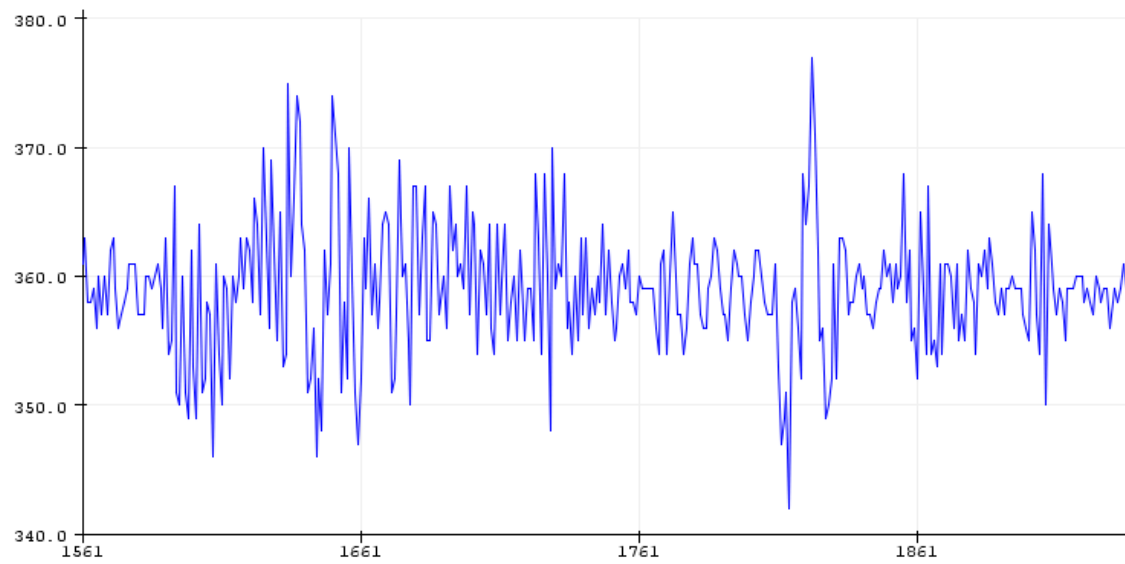
Sin modificar la soldadura nos proporciona una señal analógica en la entrada A7, que varía de cero a un valor de 3V aproximadamente, como se muestra en la imagen:



Modificación del funcionamiento

Si realizamos una pequeña soldadura como podemos ver en la imagen de la cara inferior (entre las patas delanteras de Echidna) la señal que nos proporciona ahora está centrada en $1,75V = 360$ de señal analógica, el resultado se muestra en imagen.





Pines

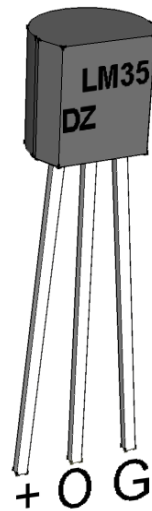
Entrada analógica A7.

Hoja de características

Sensor de temperatura LM35

Descripción

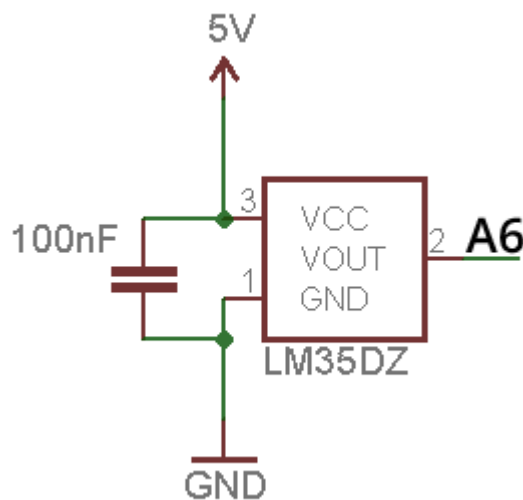
Es un sensor de temperatura calibrado cuya salida es lineal y donde cada 10 mV equivale a 1 °C



Funcionamiento

Proporciona 10 mV por °C. Teniendo en cuenta que 5V son 1024 “pasos” en la medida analógica, podemos obtener la temperatura mediante la siguiente operación:

$$\text{temperatura} = (\text{lectura Analógica} * 5.0 * 100.0) / 1024.0$$



Pines

Entrada analógica A6 (IN). Para conocer la temperatura se lee el valor del sensor y se convierte a grados: cada 10 mV es 1 °C.

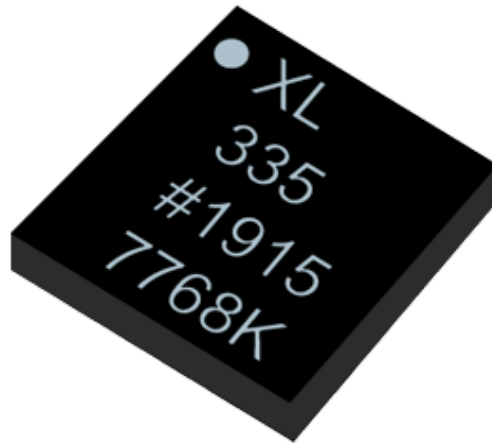
Hoja de características

Acelerómetro

Descripción

Sensor de aceleraciones basado en condensadores diferenciales dentro de una estructura micro-mecanizada.

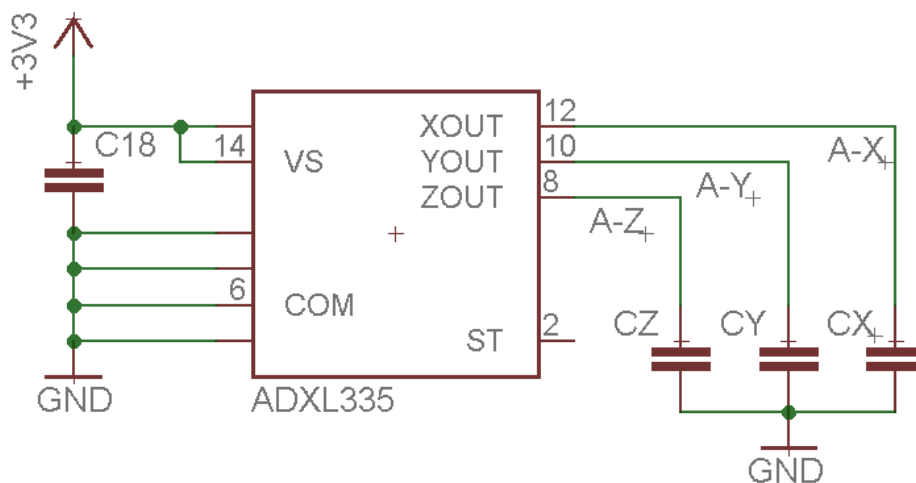
Proporciona una tensión que depende de la aceleración.



Funcionamiento

Mide la aceleración en los ejes (X, Y) en un rango de $\pm 3g$, en la posición de reposo proporciona un valor de $1,75V = 359$ (de lectura analógica)* y en sus extremos los valores $1,39V = 285^*$ - $2,10V = 428^*$.

* Debido a las tolerancias de los componentes estos valores pueden ser distintos en cada placa.



Pines

- A2 «Ace_X»: entrada analógica

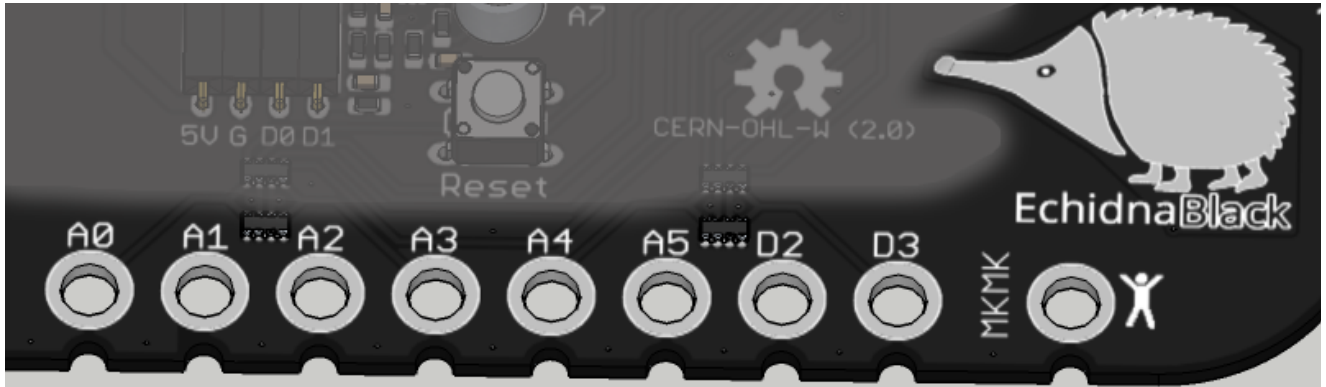
- A3 «Ace_Y»: entrada analógica

Hoja de características (anexo 6)

Conexiones MkMk

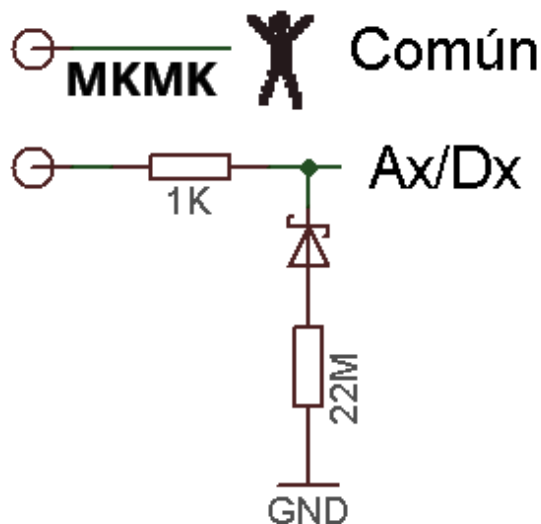
Descripción

Echidna dispone de 8 conexiones MkMk. Una conexión MkMk es un conector que permite detectar gran variedad de objetos al conectarlos entre una entrada y el común (El logo Echidna también se comporta como Común).



Funcionamiento

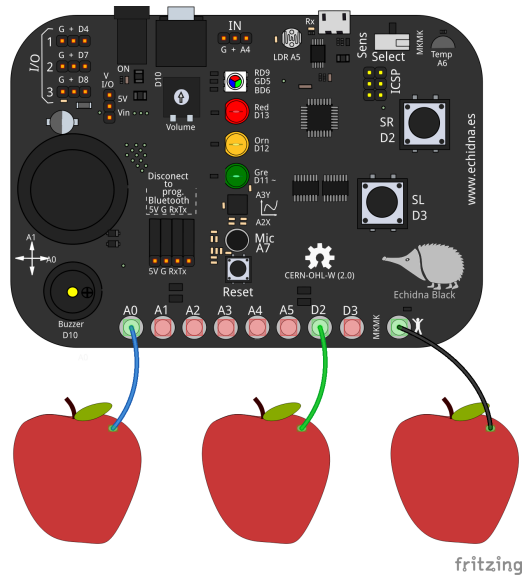
Cada conexión MkMk es parte de un divisor de tensión con una resistencia muy elevada, por lo que al conectar un elemento que conduce electricidad, aunque sea tenuemente, al cerrar el circuito con el punto común, la tensión de la entrada sube y es detectada. La tensión umbral de detección depende de cada material/persona.



Pines

- A0 MkMk0: entrada analógica
- A1 MkMk1: entrada analógica
- A2 MkMk2: entrada analógica
- A3 MkMk3: entrada analógica
- A4 MkMk4: entrada analógica

- A5 MkMk5: entrada analógica
- D2 MkMk6: entrada digital
- D3 MkMk7: entrada digital



En las entradas analógicas (A0 a A5) se compara el valor leído, de 0 a 1023, con un umbral, que puede variar según el objeto. En las digitales (D2 y D3) se lee directamente 0 o 1.

Complementos

Complementos

Estos son algunos de los **componentes** que se pueden usar para complementar la funcionalidades de **EchidnaBlack**, para conectarlos se disponen de las **I/O y la entrada analógica IN**.

Servomotor Posición

Son motores de corriente continua que permiten posicionarlo en un ángulo entre 0 y 180°.

Servomotor Continuo

Son motores de corriente continua con una reductora y electrónica de control que permiten controlar el sentido de giro.

Infrarrojos distancia

Es un sensor de distancia que proporciona una tensión según la cantidad de infrarrojo que rebota en una superficie.

Complementos conexiones MkMk

Pinzas cocodrilo y cinta conductiva para realizar las conexiones MkMk.

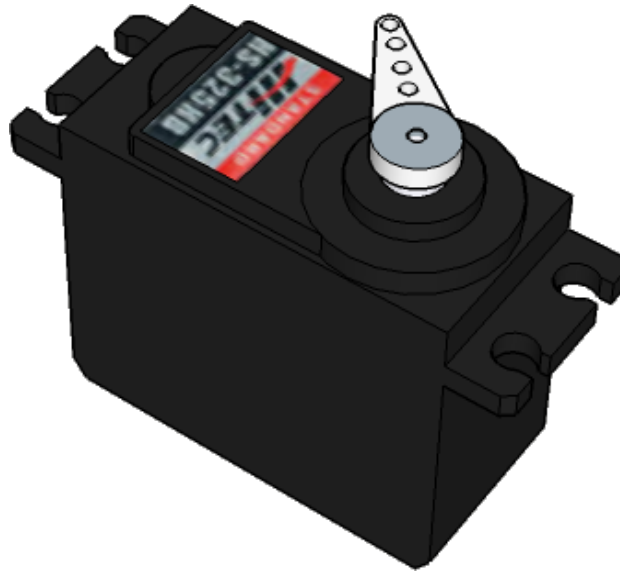
Bluetooth

Es un transceptor que permite conectar dispositivos Bluetooth a la placa Arduino.

Servomotor de posición

Descripción

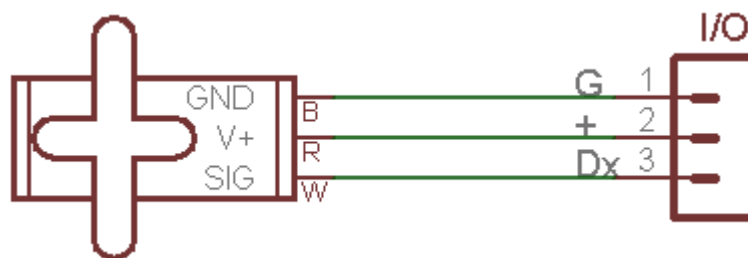
Son motores de corriente continua con una reductora y electrónica de control que permiten posicionarlo en un ángulo entre 0 y 180°.



Funcionamiento

Se controla mediante pulsos de 1-2 ms en periodos de 20 ms.

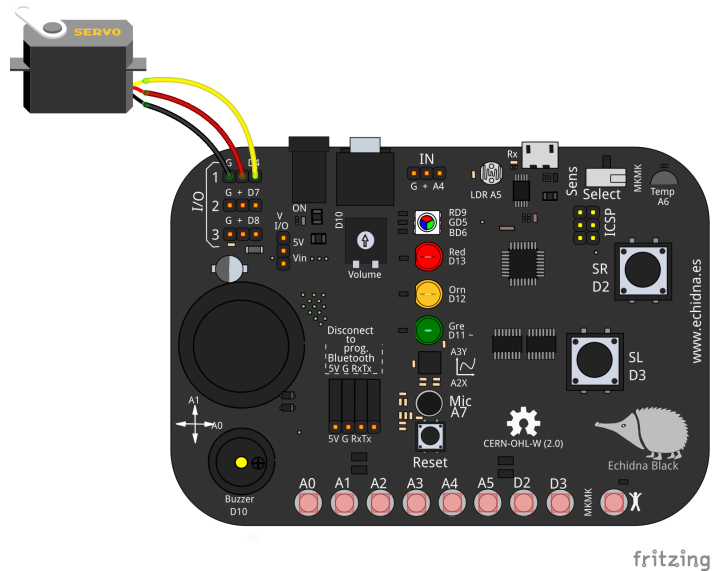
En la práctica se usa una librería que nos permite indicar directamente el ángulo.



Elementos de conexión

Para conectarlo usamos los pines I/O de entrada-salida. Se conecta directamente con sus cables.

Es aconsejable colocar el jumper de alimentación en Vin y alimentar EchidnaBlack mediante el jack de alimentación.



Pines:

1. I/O 1= PIN D4
2. I/O 2= PIN D7
3. I/O 3= PIN D8

Control

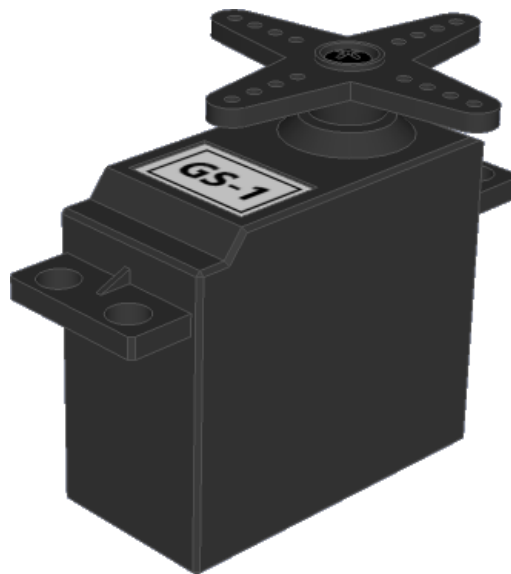
El servomotor de posición se controla indicando el ángulo, de 0 a 180 grados.

Hoja de características

Servomotor continuo

Descripción

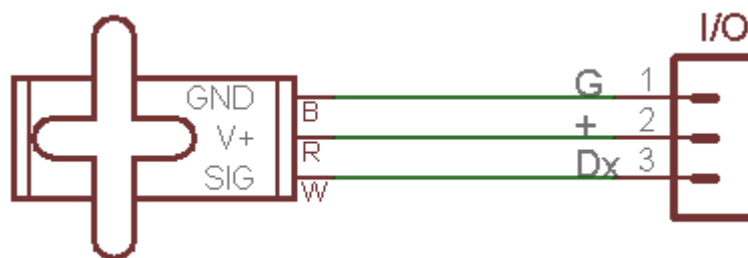
Son motores de corriente continua con una reductora y electrónica de control que permiten controlar el sentido de giro.



Funcionamiento

Se controla mediante pulsos de 1-2 ms en periodos de 20 ms.

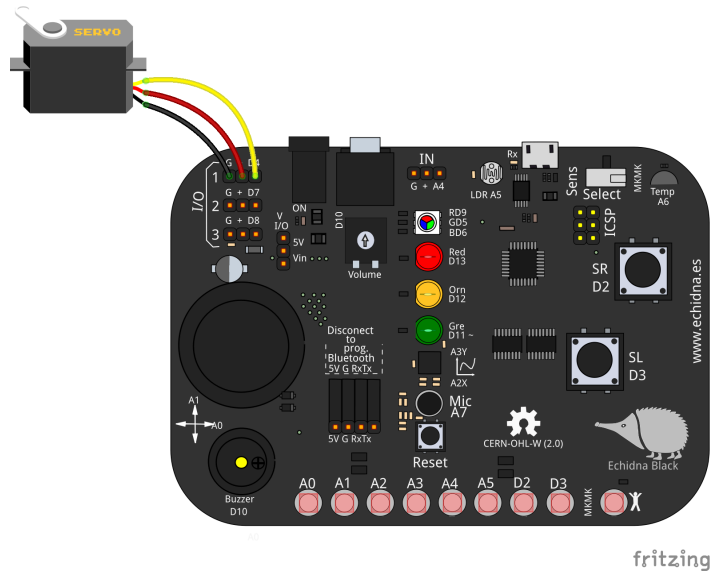
En la práctica se usa una librería que nos permite indicar directamente el sentido de giro.



Elementos de conexión

Para conectarlo usamos los pines I/O de entrada-salida. Se conecta directamente con sus cables.

Es aconsejable colocar el jumper de alimentación en Vin y alimentar EchidnaBlack mediante el jack de alimentación.



Pines:

1. I/O 1= PIN D4
2. I/O 2= PIN D7
3. I/O 3= PIN D8

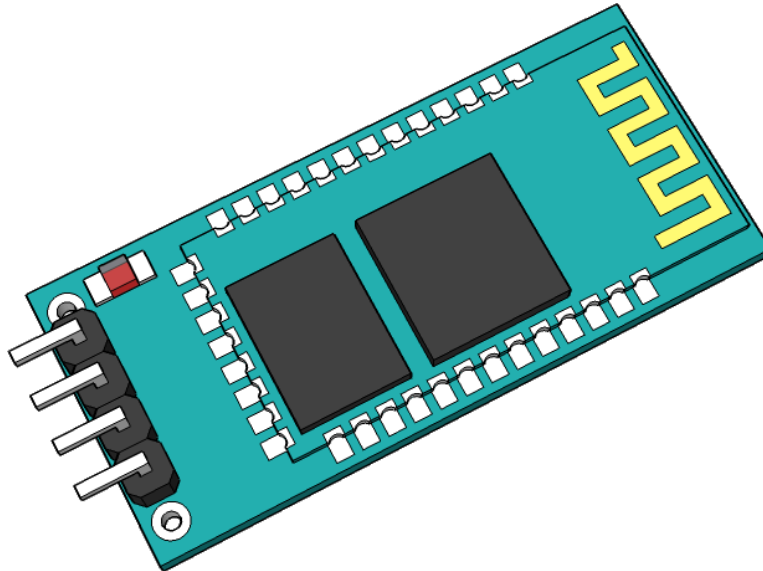
Control

El servomotor continuo se controla indicando el sentido de giro (horario o antihorario) y la velocidad.

Bluetooth

Descripción

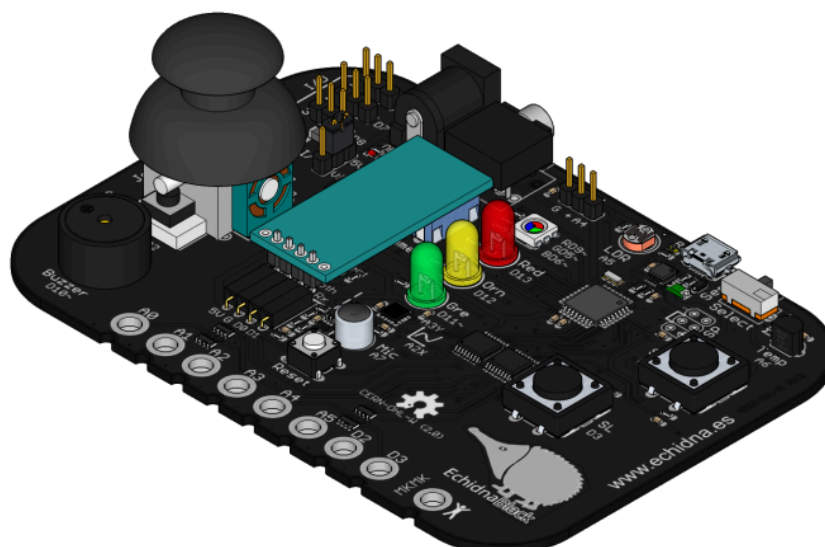
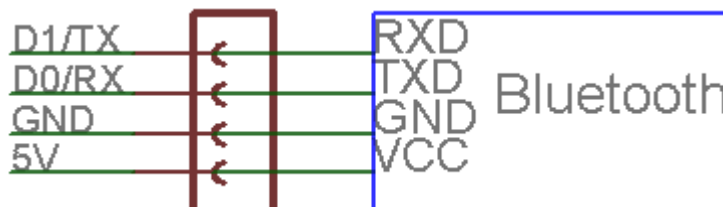
Es un transceptor que permite conectar dispositivos Bluetooth a la placa Arduino.



Funcionamiento

Envía y recibe datos a través del puerto serie vía Bluetooth

Esquema y conexión



Pines

- D0 (Rx): recepción de datos
- D1 (Tx): transmisión de datos

Se comunica a través del puerto serie Tx/Rx, es necesario programar Arduino previamente a la conexión del Bluetooth, ya que si no este bloquea la comunicación por cable.

Hoja de características

Documentación

Documentación técnica

Certificado CE

Certificado CE N°: ISETC.000520201231

Referente técnico: EBO2012089-E174

Según las directivas de compatibilidad electromagnética 2014/30/EU:

- EN 55032:2015+A11:2020; Compatibilidad electromagnética de equipos multimedia – Requisitos de emisión
- EN 55035:2017; Compatibilidad electromagnética de equipos multimedia – requisitos de inmunidad

RoHS

Los componentes y soldaduras cumplen con la normativa RoHS

Open Source Hardware Certification

EchidnaBlack está certificada como Open Source Hardware con registro **OSHW UID ES000010** por la Open Source Hardware Association (OSHW)



Licencia

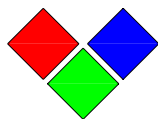
La placa Echidna Black se publica bajo licencia CERN-OHL-W 2.0.

No se permite su reproducción bajo la marca Echidna.

Anexos

Hojas de características de los componentes, en las páginas siguientes y en este orden:

1. LEDs: Hoja de características (DataSheet-LED.pdf)
2. Pulsadores: Hoja de características (DataSheet-button.pdf)
3. Joystick: Hoja de características (DataSheet-Joystick.pdf)
4. LED RGB: Hoja de características (DataSheet-LEDRGB.pdf)
5. Audio: Hoja de características (F-CM12P-LF.pdf)
6. Acelerómetro: Hoja de características (ADXL335.pdf)



深圳市昱申科技有限公司

CHINA YOUNG SUN LED TECHNOLOGY CO., LTD.

TEL: (86) 755-28079401 28079402 28079403 28079404 28079405
FAX: (86) 755-28079407 E-mail: info@100LED.com Web: www.100LED.com

Model No.: YSL-R531R3D-D2

Applications:

- Decorations
- Bill Insperctor
- Inceticial Lights
- Medical Appliance

Absolute Maximum Ratings: (Ta=25°C) .

ITEMS	Symbol	Absolute Maximum Rating	Unit
Forward Current	I _F	20	mA
Peak Forward Current	I _{FP}	30	mA
Suggestion Using Current	I _{SU}	16-18	mA
Reverse Voltage (V _R =5V)	I _R	10	uA
Power Dissipation	P _D	105	mW
Operation Temperature	T _{OPR}	-40 ~ 85	°C
Storage Temperature	T _{STG}	-40 ~ 100	°C
Lead Soldering Temperature	T _{SOL}	Max. 260°C for 3 Sec. Max. (3mm from the base of the expoxy bulb)	

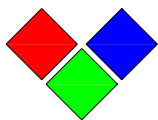
Absolute Maximum Ratings: (Ta=25°C)

ITEMS	Symbol	Test condition	Min.	Typ.	Max.	Unit
Forward Voltage	V _F	I _F =20mA	1.8	---	2.2	V
Wavelenength (nm) or TC(k)	Δ λ	I _F =20mA	620	---	625	nm
*Luminous intensity	I _v	I _F =20mA	150	---	200	mcd
50% Viewing Angle	2 θ 1/2	I _F =20mA	40	---	60	deg

Address: 5/F, Building B, Anzhilong Indl., Qinghua East Road., Longhua Town, Shenzhen CHINA. 518109

www.100LED.com

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深圳市昱申科技有限公司

CHINA YOUNG SUN LED TECHNOLOGY CO., LTD.

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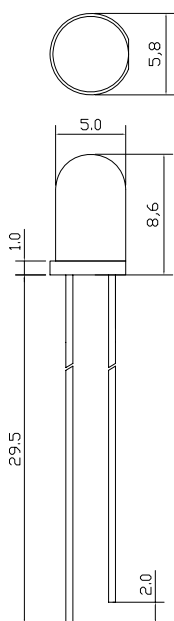
FAX: (86) 755-28079407 E-mail: info@100LED.com Web: www.100LED.com

Light Degradation in mcd: ($I_F=20mA$)

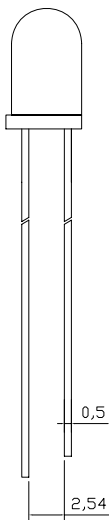
Colors	Light Degradation in mcd after Different Hours					
	216 Hrs	360 Hrs	792 Hrs	1104 Hrs	1992 Hrs	2328 Hrs
Red	1.52%	-1.22%	-3.10%	-4.68%	-5.72%	-8.27%
Yellow	-1.71%	-2.97%	-5.93%	-8.13%	-8.90%	-11.10%
Blue	3.13%	-0.33%	-3.84%	-8.23%	-21.32%	-24.92%
Green	-8.02%	-9.78%	-14.25%	-17.37%	-20.79%	-22.30%
Hours	48 Hrs	168 Hrs	336 Hrs	360Hrs	720 Hrs	1008 Hrs
Cool White	10.56%	6.72%	-2.29%	-7.68%	-17.32%	-22.48%
Pure White	13.66%	8.22%	-1.45%	-8.50%	-19.52%	-25.26%
Warm White	3.02%	-4.38%	-15.18%	-21.15%	-27.19%	-29.97%

Mechanical Dimensions:

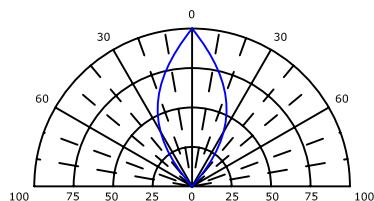
- All dimension are in mm, tolerance is $\pm 0.2mm$ unless otherwise noted
- An epoxy meniscus may extend about 1.5mm down the leads.
- Burr around bottom of epoxy may be 0.5mm Maximum



Unit: mm

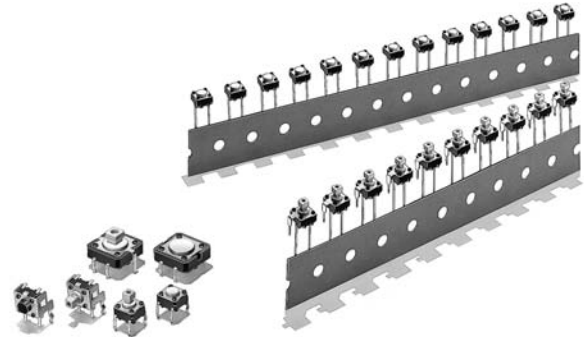


Viewing Angle Drawing



Through-hole-mounting Switches in a Wide Range of Models: 6 × 6 mm, 12 × 12 mm, Side-operated Models, Gold-plated Contacts, and Radial Tape

- Extended mechanical/electrical durability: 10 x 10⁶ operations for 12 x 12 mm type and 1 x 10⁶ operations for the 6 x 6 mm type
- Taped radial type, vertical type and high force types are available.
- Gold plated models available for increased contact reliability, resistance to corrosive gas and insulation failure prevention for ion migration in harsh environments
- B32-series Key Tops mount to models with projected plungers.

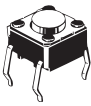
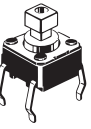
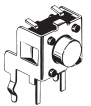


RoHS Compliant

■ List of Models

6 × 6 mm Models

Type	Contact material	Plunger	Height	Operating force (OF)	Plunger color	Bags			
						Without ground terminal	Minimum packing unit	With ground terminal	Minimum packing unit
Standard: B3F-1000 Series	Silver plated	Flat type 	4.3 mm	0.98 N {100 gf}	Ivory	B3F-1000	100 pcs	B3F-1100	100 pcs
				1.47 N {150 gf}	Yellow	B3F-1002		B3F-1102	
				2.55 N {260 gf}	Orange	B3F-1005		B3F-1105	
				4.9 N {500 gf}	Red	B3F-1006		---	
			5.0 mm	0.98 N {100 gf}	Black	B3F-1020		B3F-1120	
				1.47 N {150 gf}	Gray	B3F-1022		B3F-1122	
				2.55 N {260 gf}	Pink	B3F-1025		B3F-1125	
				4.9 N {500 gf}	Blue	B3F-1026		---	
			5.0 mm (7.5-mm pitch)	0.98 N {100 gf}	Black	---		B3F-1110	
			7.0 mm	0.98 N {100 gf}	Black	B3F-1060		---	
				1.47 N {150 gf}	Yellow	B3F-1062		---	
				0.98 N {100 gf}	Black	B3F-1070		---	
				1.47 N {150 gf}	Yellow	B3F-1072		---	
			9.5 mm	2.55 N {260 gf}	Orange	B3F-1075		---	
				0.98 N {100 gf}	Ivory	B3F-1050		B3F-1150	
				1.47 N {150 gf}	Yellow	B3F-1052		B3F-1152	
				2.55 N {260 gf}	Orange	B3F-1055		B3F-1155	
				4.9 N {500 gf}	Red	B3F-1056		---	
		Projected type 	7.3 mm	0.98 N {100 gf}	Ivory	B3F-1050		B3F-1150	
				1.47 N {150 gf}	Yellow	B3F-1052		B3F-1152	
				2.55 N {260 gf}	Orange	B3F-1055		B3F-1155	
				4.9 N {500 gf}	Red	B3F-1056		---	

Type	Contact material	Plunger	Height	Operating force (OF)	Plunger color	Bags			
						Without ground terminal	Minimum packing unit	With ground terminal	Minimum packing unit
Side-operated: B3F-3000 Series	Silver plated	 Flat type	3.15 mm	0.98 N {100 gf}	Ivory	---	100 pcs	B3F-3100	100 pcs
				1.47 N {150 gf}	Yellow	---		B3F-3102	
				2.55 N {260 gf}	Orange	---		B3F-3105	
			3.85 mm	0.98 N {100 gf}	Black	---		B3F-3120	
				1.47 N {150 gf}	Gray	---		B3F-3122	
				2.55 N {260 gf}	Pink	---		B3F-3125	
		 Projected type	6.15 mm	0.98 N {100 gf}	Ivory	---		B3F-3150	
				1.47 N {150 gf}	Yellow	---		B3F-3152	
				2.55 N {260 gf}	Orange	---		B3F-3155	
High-reliability gold-plated: B3F-1000-G Series	Gold plated	 Flat type	4.3 mm	1.76 N {180 gf}	Yellow	B3F-1002-G		B3F-1102-G	
			5.0 mm		Gray	B3F-1022-G		B3F-1122-G	
			7.0 mm		Yellow	B3F-1062-G (see note)		---	
			9.5 mm		Yellow	B3F-1072-G (see note)		---	
Side-operated with highly reliable gold-plated: B3F-3000-G Series	Gold plated	 Projected type	7.3 mm		Yellow	B3F-1052-G		---	
		 Flat type	3.85 mm		Gray	---		B3F-3122-G (see note)	

Note: Bulk Packaged, 100 Switches per bag. Order in multiples of the package quantity.

12 × 12 mm Models

Type	Contact material	Plunger (or LED color)	Height	Operating force (OF)	Plunger color	Bags			
						Without ground terminal	Minimum packing unit	With ground terminal	Minimum packing unit
Standard: B3F-4000 Series	Silver plated	Flat type	4.3 mm	1.27 N {130 gf}	Ivory	B3F-4000	100 pcs	B3F-4100	100 pcs
				2.55 N {260 gf}	Yellow	B3F-4005		B3F-4105	
		Projected type	7.3 mm	1.27 N {130 gf}	Ivory	B3F-4050		B3F-4150	
				2.55 N {260 gf}	Yellow	B3F-4055		B3F-4155	
Long durability: B3F-5000 Series	Silver plated	Flat type	4.3 mm	1.27 N {130 gf}	Blue	B3F-5000		B3F-5100	
		Projected type	7.3 mm		Blue	B3F-5050		B3F-5150	
High reliability gold-plated: B3F-5001 Series	Gold plated	Flat type	4.3 mm	1.27 N {130 gf}	Blue	B3F-5001		B3F-5101	
		Projected type	7.3 mm		Blue	B3F-5051		B3F-5151	

Note: Bulk Packaged, 100 switches per bag. Order in multiples of the package quantity.

6 × 6 mm Radial Models (Taping Specifications)

Type	Contact material	Plunger	Height	Operating force (OF)	Plunger color	Taped Radial			
						Without ground terminal	Minimum packing unit	With ground terminal	Minimum packing unit
Taped Radial: B3F-6000 Series	Silver plated	Flat type	4.3 mm	0.98 N {100 gf}	Ivory	B3F-6000	1,000 pcs	B3F-6100	1,000 pcs
				1.47 N {150 gf}	Yellow	B3F-6002		B3F-6102	
			5.0 mm	0.98 N {100 gf}	Black	B3F-6020		B3F-6120	
				1.47 N {150 gf}	Gray	B3F-6022		B3F-6122	
		Projected type	7.3 mm	0.98 N {100 gf}	Ivory	B3F-6050		B3F-6150	
				1.47 N {150 gf}	Yellow	B3F-6052		B3F-6152	

Note: The switches are tape packaged in units of 1,000 per package. Order in multiples of the package size. Switches are not sold individually.

■ Ratings/Characteristics

Rating (resistive load)	1 to 50 mA, 3 to 24 VDC (B3F-G: 100 μ A to 50 mA, 3 to 24 VDC)
Minimum applicable load (reference value)	10 μ A at 1 VDC (resistive load)
Ambient operating temperature	-25°C to +70°C at 60%RH max. (with no icing or condensation)
Ambient operating humidity	35% to 85% (at +5 to +35°C)
Contact form	SPST-NO
Contact resistance (initial value)	100 m Ω max.
Insulation resistance	100 M Ω min. (at 250 VDC with insulation tester)
Dielectric strength	500 VAC, 50/60 Hz for 1 min
Bounce time	5 ms max.
Vibration resistance	Malfunction: 10 to 55 Hz, 1.5 mm double amplitude
Shock resistance	Destruction: 1,000 m/s ² {approx. 100G} max. Malfunction: 100 m/s ² {approx. 10G} max.
Durability	B3F-1000, B3F-3000, B3F-6000: 1,000,000 operations min (OF: 0.98 N {100 gf}) (B3F-1070: 500,000 operations min) 300,000 operations min (OF: 1.47 N {150 gf}) 100,000 operations min (OF: 2.55 N {260 gf}) 50,000 operations min (OF: 4.9 N {500 gf}) B3F-4000: 3,000,000 operations min (OF: 1.27 N {130 gf}) 1,000,000 operations min (OF: 2.55 N {260 gf}) B3F-5000/5001: 10,000,000 operations min. B3F-G: 300,000 operations min.
Weight	6 $\times$ 6 mm models: approx. 0.25 g 12 $\times$ 12 mm models (standard types): approx. 0.85 g Radial models: approx. 0.25 g
Degree of protection	IEC IP00
Washing	Not possible

■ Operating Characteristics

6 $\times$ 6 mm Models

Operating force (OF)	B3F-1000, B3F-3000, B3F-6000				B3F-G
	0.98 N	1.47 N	2.55 N	4.9 N	1.76 N
Item	B3F-1□□0 B3F-3□□0 B3F-6□□0	B3F-1□□2 B3F-3□□2 B3F-6□□2	B3F-1□□5 B3F-3□□5	B3F-10□6	B3F-1□□2-G B3F-3□□2-G
Operating force (OF)	0.98 $\pm$ 0.29 N {100 $\pm$ 30 gf}	1.47 $\pm$ 0.49 N {150 $\pm$ 50 gf}	2.55 $\pm$ 0.69 N {260 $\pm$ 70 gf}	4.9 $\pm$ 1.47 N {500 $\pm$ 150 gf}	1.76 $\pm$ 0.49 N {180 $\pm$ 50 gf}
Releasing force (RF)	0.2 N {20 gf} min.	0.49 N {50 gf} min.	0.49 N {50 gf} min.	0.7 N {70 gf} min.	0.49 N {50 gf} min.
Pretravel (PT)	0.25 ^{+0.2} / _{-0.1} mm				0.25 ^{+0.2} / _{-0.1} mm

12 $\times$ 12 mm Models

Operating force (OF)	B3F-4000, B3F-5000, B3F-5001	
	1.27 N	2.55 N
Item	B3F-4□□0 B3F-5□□0 B3F-5□□1	B3F-4□□5
Operating force (OF)	1.27 $\pm$ 0.49 N {130 $\pm$ 50 gf}	2.55 $\pm$ 0.69 N {260 $\pm$ 70 gf}
Releasing force (RF)	0.29 N {30 gf} min.	0.49 N {50 gf} min.
Pretravel (PT)	0.3 ^{+0.2} / _{-0.1} mm	

■ Dimensions (Unit: mm)

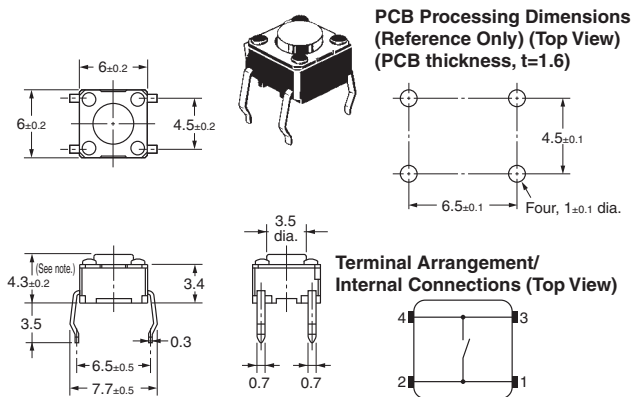
Note: The numbers used for terminals in the following graphics are indicated in the "Bottom View" diagram below. In this diagram, the Switch is rotated so that the terminals are on the right and left-hand sides, and the OMRON logo appears the right way up. (Except Side-operated and Radial Models)



6 × 6 mm Models

Standard, Flat Plunger Type (without Ground Terminal)

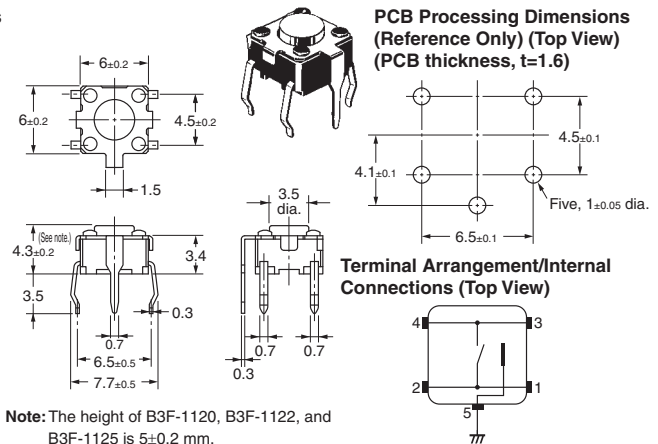
B3F-1000, B3F-1002, B3F-1005, B3F-1006
B3F-1020 (See note.), B3F-1022 (See note.),
B3F-1025 (See note.), B3F-1026 (See note.)
B3F-1002-G, B3F-1022-G (See note.)



Note: The height of B3F-1020, B3F-1022, B3F-1025, and B3F-1026 is 5 ± 0.2 mm.

Standard, Flat Plunger Type (with Ground Terminal)

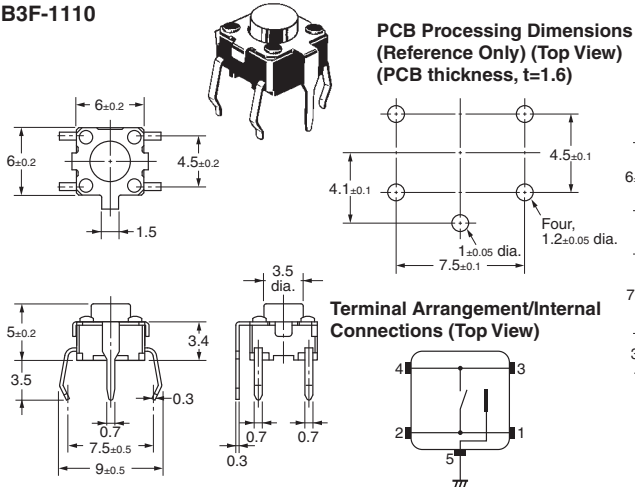
B3F-1100, B3F-1102, B3F-1105
B3F-1120 (See note.), B3F-1122 (See note.)
B3F-1125 (See note.)
B3F-1102-G, B3F-1122-G (See note.)



Note: The height of B3F-1120, B3F-1122, and B3F-1125 is 5 ± 0.2 mm.

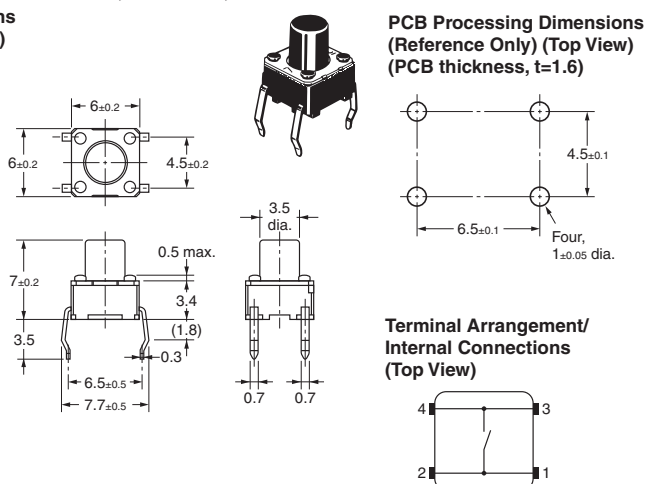
Standard, Flat Plunger Type (with Ground Terminal, Pitch: 7.5 mm)

B3F-1110



Standard, Flat Plunger Type (without Ground Terminal)

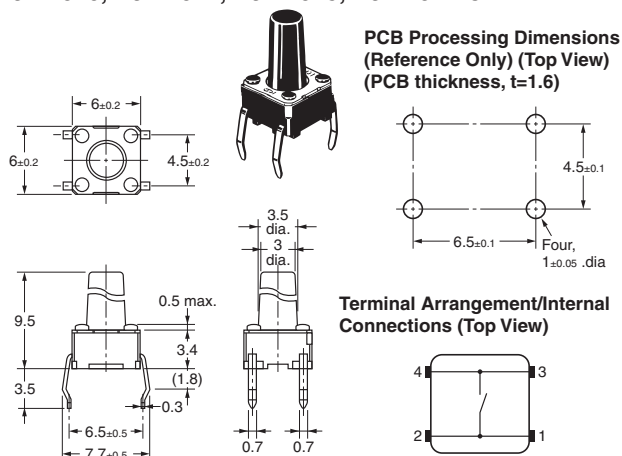
B3F-1060, B3F-1062, B3F-1062-G



Note: Unless otherwise specified, a tolerance of ±0.4 mm applies to all dimensions. No terminal numbers are indicated on the Switches.

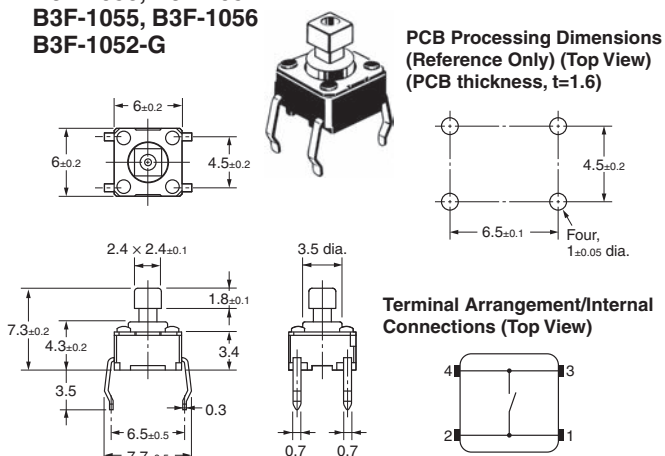
Standard, Flat Plunger Type (without Ground Terminal)

B3F-1070, B3F-1072, B3F-1075, B3F-1072-G



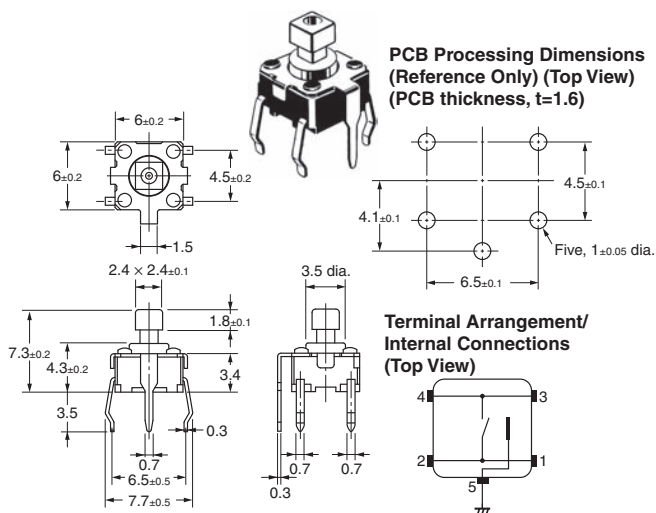
Standard, Projected Plunger Type (without Ground Terminal)

B3F-1050, B3F-1052
B3F-1055, B3F-1056
B3F-1052-G



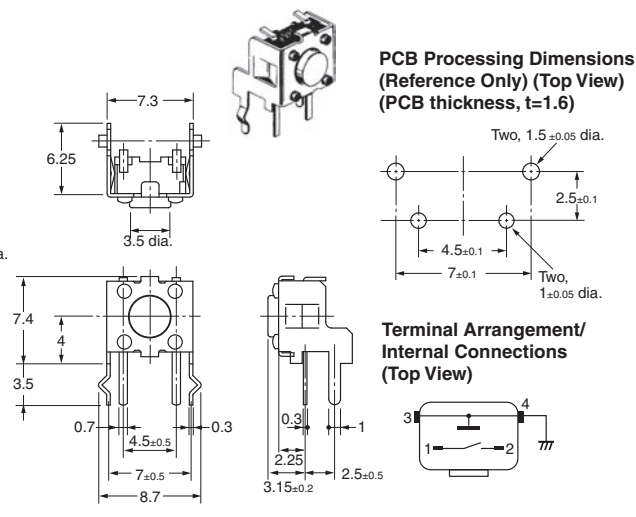
Standard, Projected Plunger Type (with Ground Terminal)

B3F-1150, B3F-1152, B3F-1155



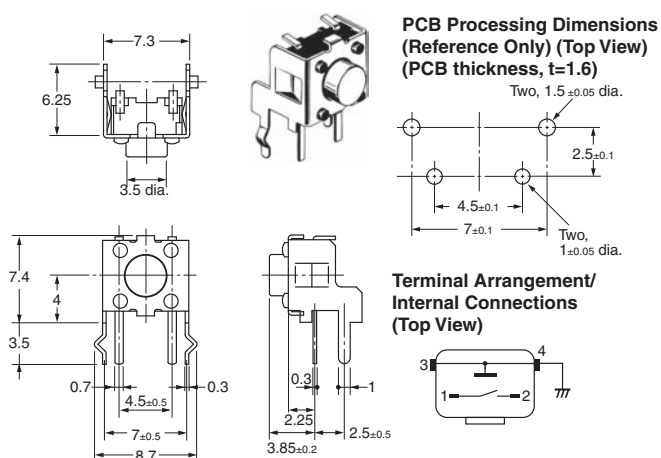
Side-operated, Flat Plunger Type

B3F-3100, B3F-3102, B3F-3105



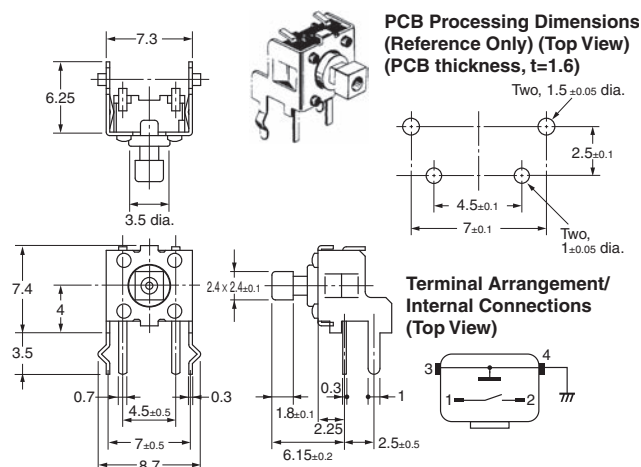
Side-operated, Flat Plunger Type (Height: 3.85 mm)

B3F-3120, B3F-3122, B3F-3125, B3F-3122-G



Side-operated, Projected Plunger Type

B3F-3150, B3F-3152, B3F-3155

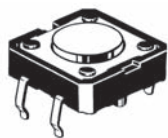


Note: Unless otherwise specified, a tolerance of ± 0.4 mm applies to all dimensions. No terminal numbers are indicated on the Switches.

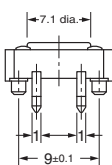
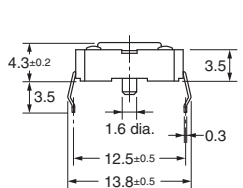
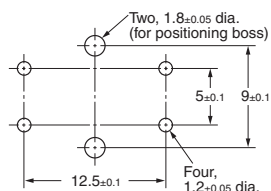
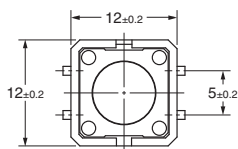
12 × 12 mm Models

Standard, Long-durability,
and High-reliability Models
Flat Plunger Type
(without Ground Terminal)

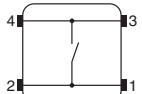
B3F-4000, B3F-4005,
B3F-5000, B3F-5001



PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, $t=1.6$)

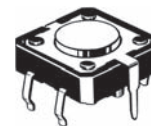


Terminal Arrangement/
Internal Connections
(Top View)

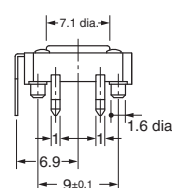
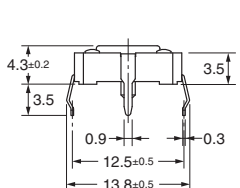
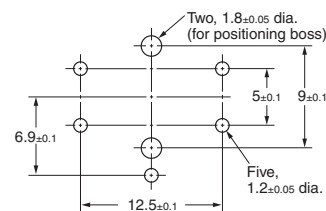
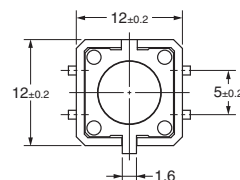


Standard, Long-durability,
and High-reliability Models
Flat Plunger Type
(with Ground Terminal)

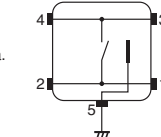
B3F-4100, B3F-4105,
B3F-5100, B3F-5101



PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, $t=1.6$)

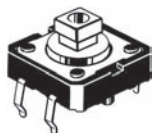


Terminal Arrangement/
Internal Connections
(Top View)

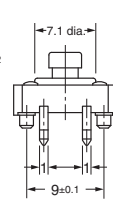
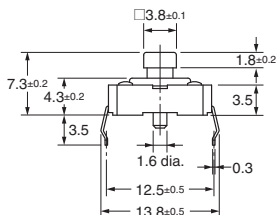
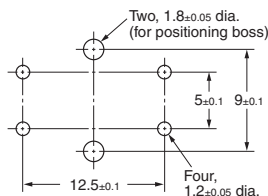
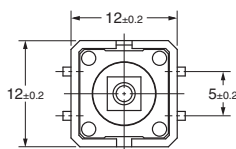


Standard, Long-durability,
and High-reliability Models
Projected Plunger Type
(without Ground Terminal)

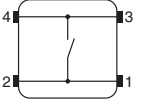
B3F-4050, B3F-4055,
B3F-5050, B3F-5051



PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, $t=1.6$)

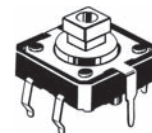


Terminal Arrangement/
Internal Connections
(Top View)

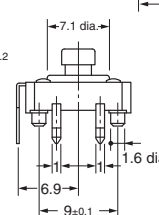
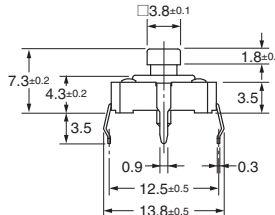
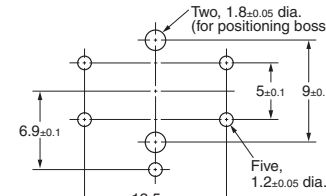
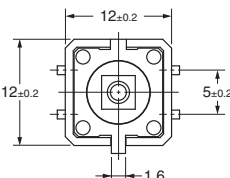


Standard, Long-durability,
and High-reliability Models
Projected Plunger Type
(with Ground Terminal)

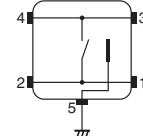
B3F-4150, B3F-4155,
B3F-5150, B3F-5151



PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, $t=1.6$)

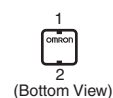


Terminal Arrangement/
Internal Connections
(Top View)



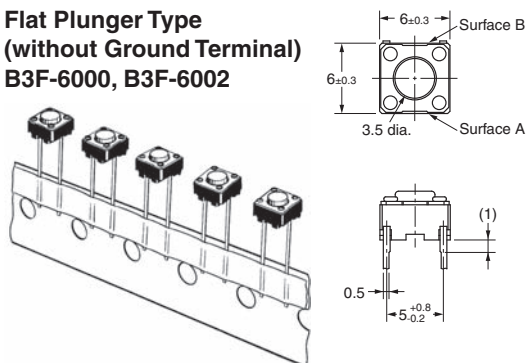
Note: Unless otherwise specified, a tolerance of ± 0.4 mm applies to all dimensions.

Note: The numbers used for terminals in the following graphics are indicated in the "Bottom View" diagram below. In this diagram, the Switch is rotated so that the terminals are on the right and left-hand sides, and the OMRON logo appears the right way up.



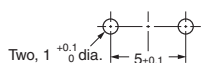
6 mm × 6 mm Radial Types (Taping Specifications): Sold in Units of 1,000 Switches

Flat Plunger Type (without Ground Terminal) B3F-6000, B3F-6002

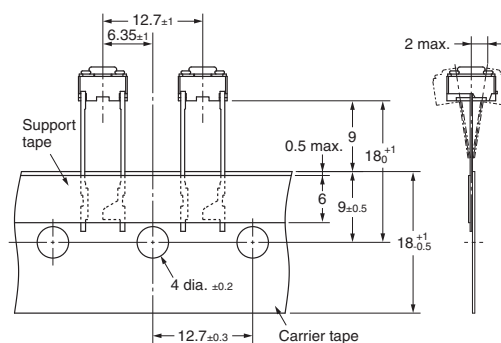
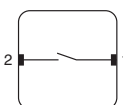


Note: The tape is random between surface A and surface B.

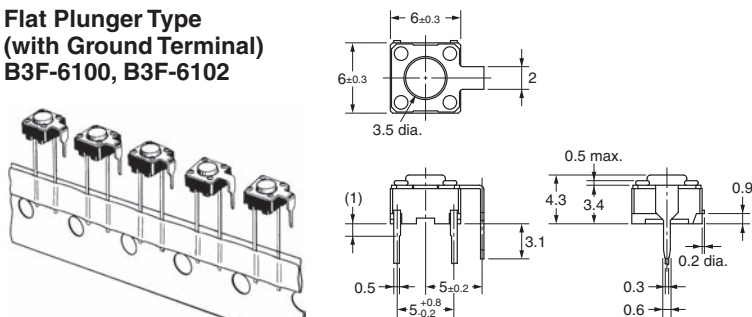
**PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, t=1.6)**



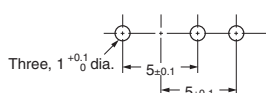
**Terminal Arrangement
/Internal Connections
(Top View)**



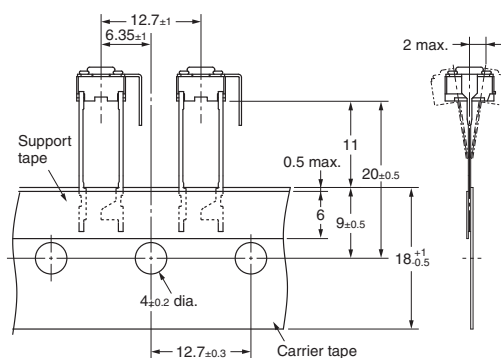
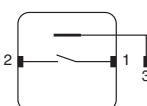
Flat Plunger Type (with Ground Terminal) B3F-6100, B3F-6102



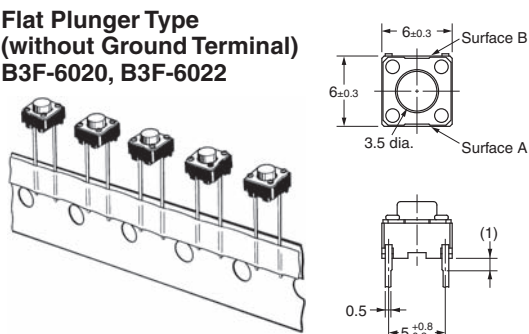
**PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, t=1.6)**



**Terminal Arrangement
/Internal Connections
(Top View)**



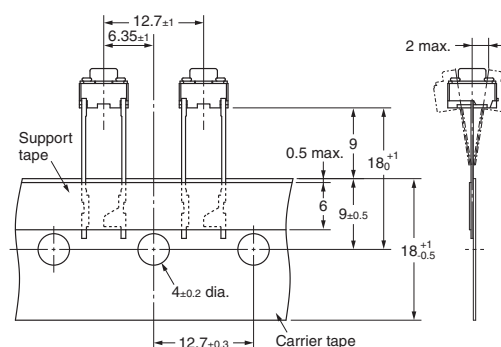
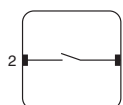
Flat Plunger Type (without Ground Terminal) B3F-6020, B3F-6022



**PCB Processing Dimensions
(Reference Only) (Top View)
(PCB thickness, t=1.6)**

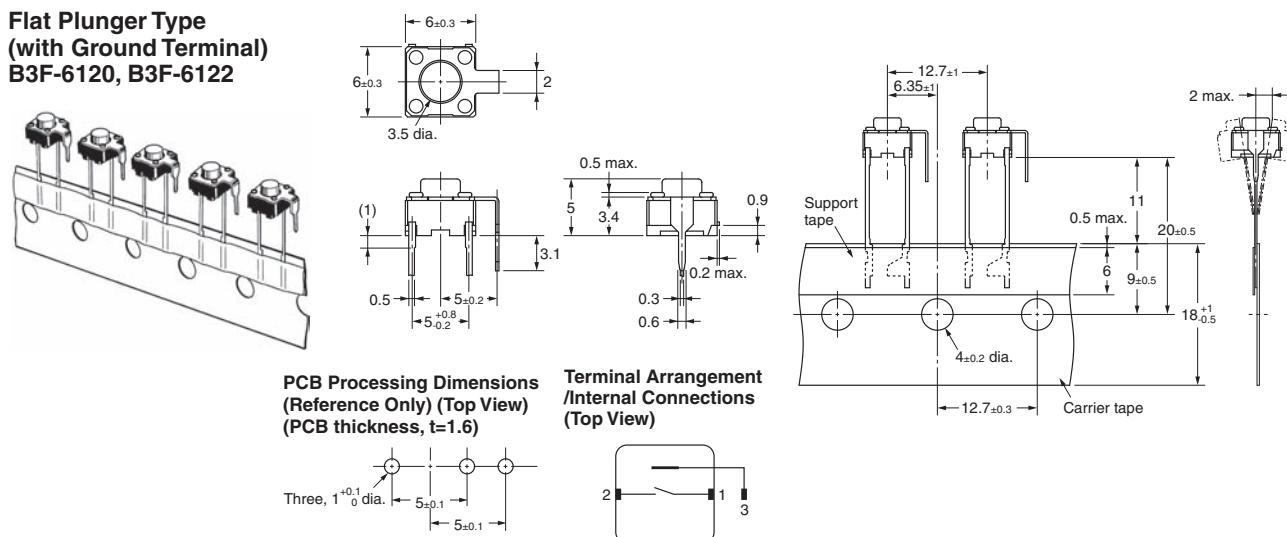


**Terminal Arrangement
/Internal Connections
(Top View)**

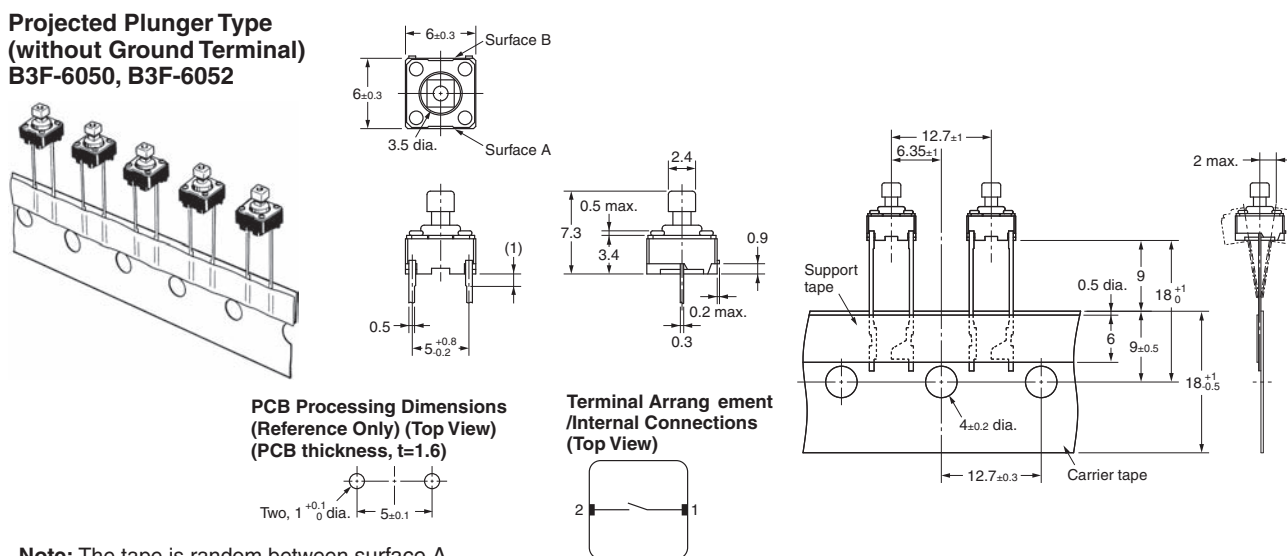


Note: Unless otherwise specified, a tolerance of ± 0.4 mm applies to all dimensions. No terminal numbers are indicated on the Switches.

Flat Plunger Type (with Ground Terminal) B3F-6120, B3F-6122

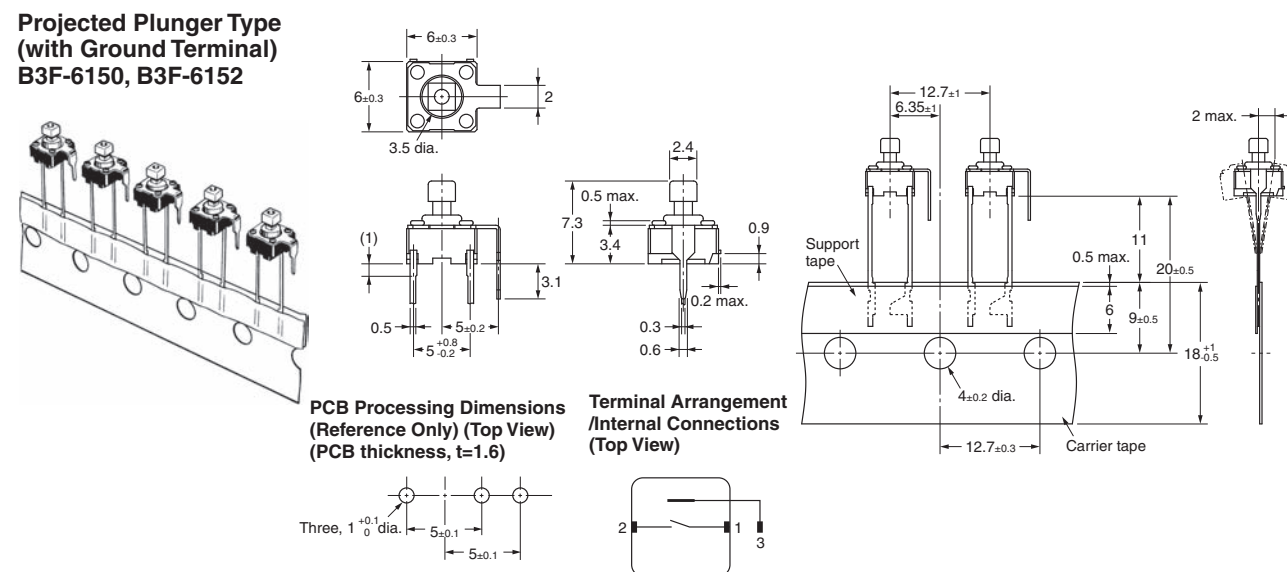


Projected Plunger Type (without Ground Terminal) B3F-6050, B3F-6052



Note: The tape is random between surface A and surface B.

Projected Plunger Type (with Ground Terminal) B3F-6150, B3F-6152



Note: Unless otherwise specified, a tolerance of ± 0.4 mm applies to all dimensions. No terminal numbers are indicated on the Switches.

■ Key Tops

B32-series Key Tops are available for projected plungers. Refer to the Datasheet of B32 for details.

■ Precautions

Be sure to read the safety precautions common to all Tactile Switches for correct use.

- Application examples provided in this document are for reference only. In actual applications, confirm equipment functions and safety before using the product.
- Consult your OMRON representative before using the product under conditions which are not described in the manual or applying the product to nuclear control systems, railroad systems, aviation systems, vehicles, combustion systems, medical equipment, amusement machines, safety equipment, and other systems or equipment that may have a serious influence on lives and property if used improperly. Make sure that the ratings and performance characteristics of the product provide a margin of safety for the system or equipment, and be sure to provide the system or equipment with double safety mechanisms.

Note: Do not use this document to operate the Unit.

OMRON Corporation

Electronic and Mechanical Components Company

Contact: www.omron.com/ecb

Cat. No. A070-E1-08
1014(0207)(O)



Feature

1. Superior operational feeling free from gaps in operating directions
Ideal for finger-tip operation.
2. Integrated tact switch for selection contributes to the improvement of user-friendliness of the set.
3. Long Life Type Available.

FJN10K Series Joystick Potentiometer

Electrical characteristics

- Total Resistance.....5K Ω ~100K Ω
- Total Resistance Tolerance..... $\pm 20\%$
- Resistance Taper.....B taper
- Rated Power.....Taper B:0.0125W
- Withstand Voltage.....1 minute at AC 250V
- Insulation Resistance.....More than 100M Ω at DC 250V for 1 minute
- Max. Operating Voltage.....Taper B:AC 50V or DC 5.0V

Mechanical characteristics

- Figure of Lever Operation.....Circular Operating
- Stopper Strength3.1Kgf.cm Min 3Sec.
- Operating Angle of Lever..... $60^\circ \pm 6^\circ$
- Push-Pull Strength.....Push/Pull:5.0Kgf Min 3Sec.
- Operating Force.....160 $\pm$ 40gf.cm
- Accuracy of Rest Position..... $\pm 5^\circ$

Durability

- Operating Life.....Standard Life: 1Million Cycles min
Long Life: 2 Million Cycles min
- Operating Temperature Range..... $-10^\circ\text{C} \sim +70^\circ\text{C}$

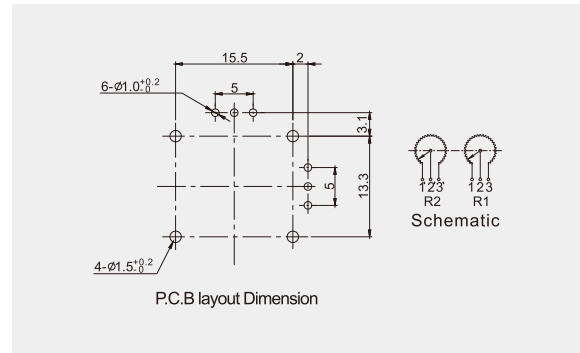
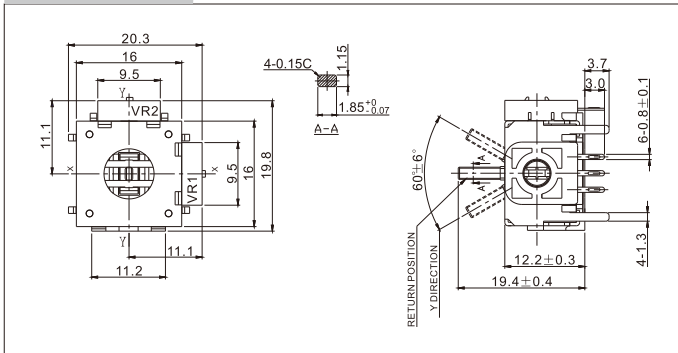
Tact Switch specification

- Ratings.....12 V DC, 50 mA
- Operating Force.....820 $\pm$ 260gf
- Travel.....0.5+0.5/-0.4mm
- Operating Life.....100,000Cycles min

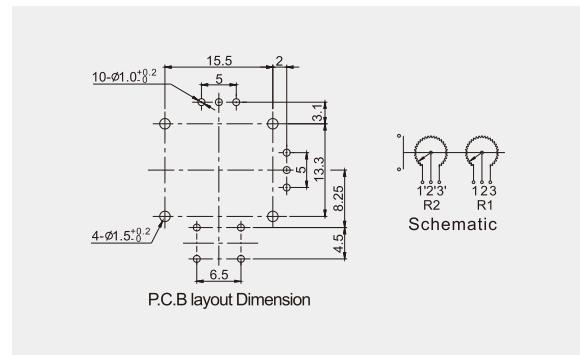
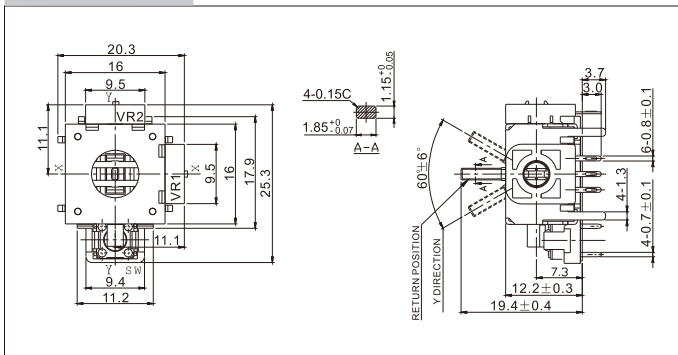
HOW TO ORDER

FJN10K-		N	1	B	100K	29	8
Model Series				Total Resistance		Electric Effective Angle	
Code	Switch Option			5K		See Note1	
N	No Switch			10K			
S	With Tact Switch			20K			
				50K			
				100K			

FJN10K-N1 Metal Shaft, No Switch, PC Pins



FJN10K-S1 Metal Shaft with Tact Switch, PC Pins



Cree® PLCC6 3-in-1 SMD LED CLP6C-FKB



PRODUCT DESCRIPTION

Cree PLCC full-color LEDs offer high-intensity light output and a wide viewing angle in an industry-standard package. Designed to work in a wide array of environmental conditions, Cree PLCC full-color LEDs are suited for indoor video screen, decorative lighting and amusement applications.

FEATURES

- Size (mm):6.0 x 5.0
- Dominant Wavelength:
Red (619 - 624nm)
Green (520 - 540nm)
Blue (460 - 480nm)
- Luminous Intensity (mcd)
Red (560 - 1120)
Green (1120 - 2240)
Blue (280 - 560)
- Lead-Free
- RoHS Compliant

APPLICATIONS

- Full-Color Video Screen
- Decorative lighting
- Amusement

ABSOLUTE MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$)

Items	Symbol	Absolute Maximum Rating			Unit
		R	G	B	
Forward Current ^{Note 1}	I_F	50	50	50	mA
Peak Forward Current ^{Note 2}	I_{FP}	200	100	100	mA
Reverse Voltage	V_R	5	5	5	V
Power Dissipation	P_D	130	200	200	mW
Operation Temperature	T_{opr}	-40 ~ +100			$^\circ\text{C}$
Storage Temperature	T_{stg}	-40 ~ +100			$^\circ\text{C}$
Junction Temperature	T_J	110	110	110	$^\circ\text{C}$
Junction/ambient 1 chip on	R_{THJA}	450	400	450	$^\circ\text{C/W}$
Junction/ambient 3 chips on	R_{THJA}	650	580	680	$^\circ\text{C/W}$
Junction/solder point 1 chip on	R_{THJS}	300	280	300	$^\circ\text{C/W}$
Junction/solder point 3 chips on	R_{THJS}	450	430	480	$^\circ\text{C/W}$
Electrostatic Discharge Classification(MIL-STD-883E)	ESD	1000 V			

Note: 1.Single-color light.
2.Pulse width ≤ 0.1 msec, duty $\leq 1/10$.

TYPICAL ELECTRICAL & OPTICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$)

Characteristics	Condition	Symbol	Values			Unit
			R	G	B	
Dominant Wavelength	$I_F = 20$ mA	λ_{DOM}	619~624	520~540	460~480	nm
Spectral bandwidth at 50% I_{REL} max	$I_F = 20$ mA	$\Delta \lambda$	24	38	28	nm
Forward Voltage	$I_F = 20$ mA	$V_{F(avg)}$	2.0	3.2	3.2	V
		$V_{F(max)}$	2.6	4.0	4.0	V
Luminous Intensity	$I_F = 20$ mA	$I_{V(min)}$	560	1120	280	mcd
		$I_{V(avg)}$	700	1600	400	mcd
Reverse Current (max)	$V_R = 5$ V	I_R	10	10	10	μA

INTENSITY BIN LIMIT ($I_F = 20 \text{ mA}$)

Red			Green			Blue		
Bin Code	Min.(mcd)	Max.(mcd)	Bin Code	Min.(mcd)	Max.(mcd)	Bin Code	Min.(mcd)	Max.(mcd)
K	560	710	P	1120	1400	G	280	355
M	710	900	Q	1400	1800	H	355	450
N	900	1120	R	1800	2240	J	450	560

Tolerance of measurement of luminous intensity is $\pm 10\%$.

COLOR BIN LIMIT ($I_F = 20 \text{ mA}$)

Red			Green			Blue		
Bin Code	Min.(nm)	Max.(nm)	Bin Code	Min.(nm)	Max.(nm)	Bin Code	Min.(nm)	Max.(nm)
RB	619	624	G7	520	525	B3	460	465
			G8	525	530	B4	465	470
			G9	530	535	B5	470	475
			Ga	535	540	B6	475	480

Tolerance of measurement of dominant wavelength is $\pm 1 \text{ nm}$.

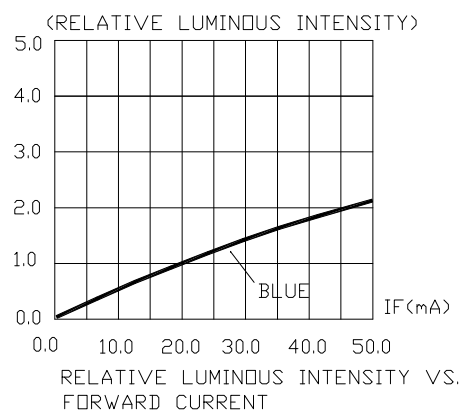
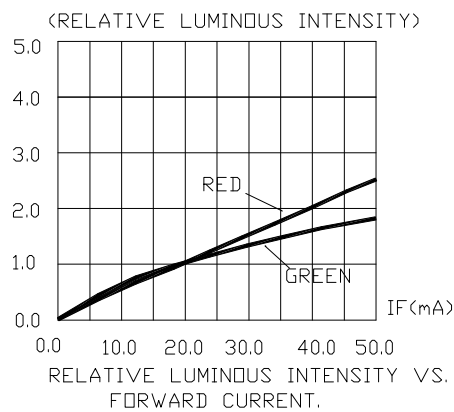
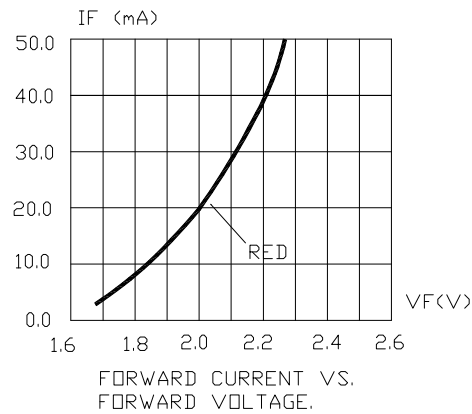
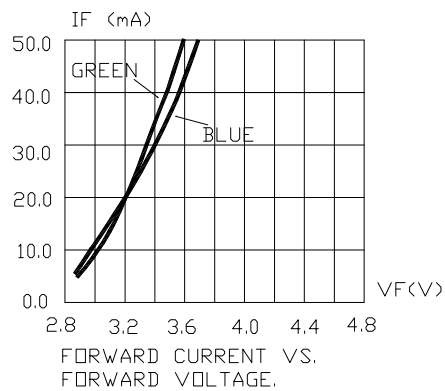
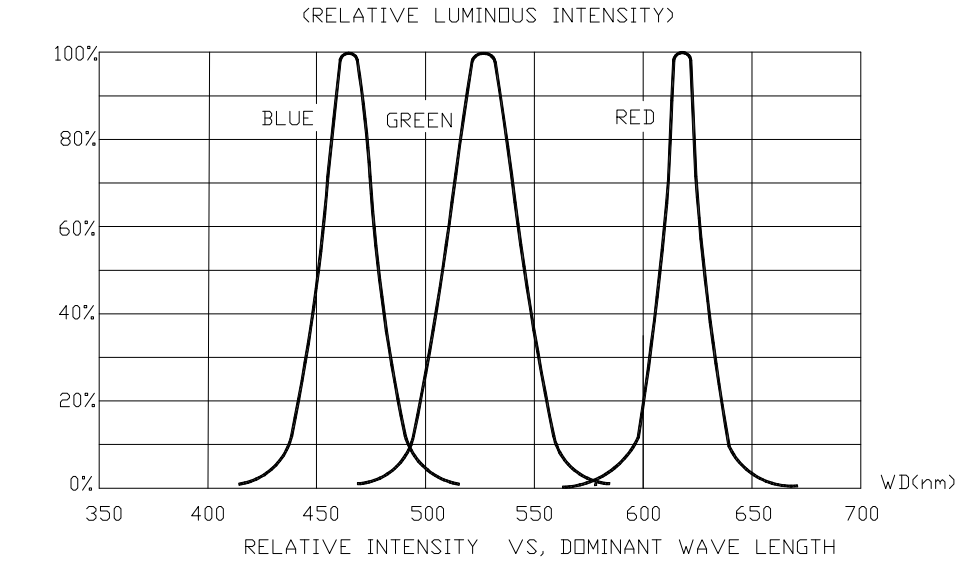
ORDER CODE TABLE*

Kit Number	Color	Luminous Intensity (mcd)		Dominant Wavelength (nm)				Package
		Min.	Max.	Color Bin	Min. (nm)	Color Bin	Max. (nm)	
CLP6C-FKB-CKNPRGJBB7a363	Red	560	1120	RB	619	RB	624	Reel
	Green	1120	2240	G7	520	Ga	540	Reel
	Blue	280	560	B3	460	B6	480	Reel
CLP6C-FKB-CK1P1G1BB7R3R3	Red	Any 1 Intensity bin from K(560) - N(1120)		RB	619	RB	624	Reel
	Green	Any 1 Intensity bin from P(1120) - R(2240)		Any 1 hue bin from G7(520) - Ga(540)				Reel
	Blue	Any 1 Intensity bin from G(280) - J(560)		Any 1 hue bin from B3(460) - B6(480)				Reel
CLP6C-FKB-CM1Q1H1BB7R3R3	Red	Any 1 Intensity bin from M(710) - N(1120)		RB	619	RB	624	Reel
	Green	Any 1 Intensity bin from Q(1400) - R(2240)		Any 1 hue bin from G7(520) - Ga(540)				Reel
	Blue	Any 1 Intensity bin from H(355) - J(560)		Any 1 hue bin from B3(460) - B6(480)				Reel

Notes:

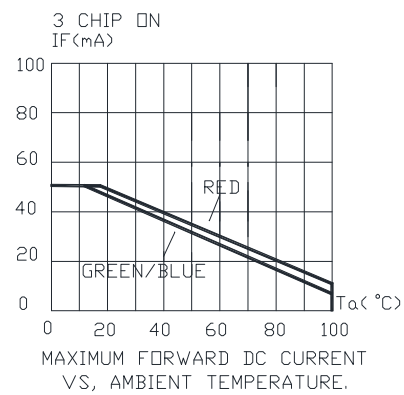
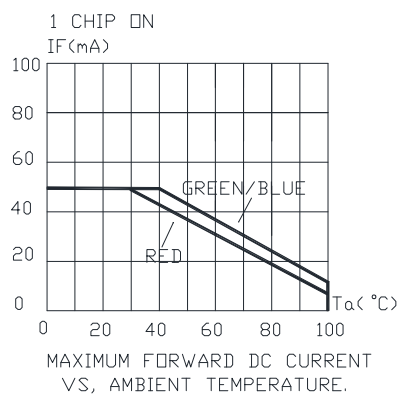
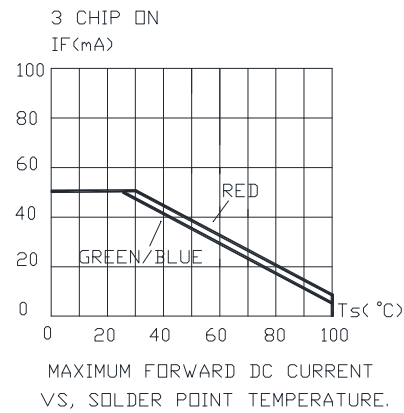
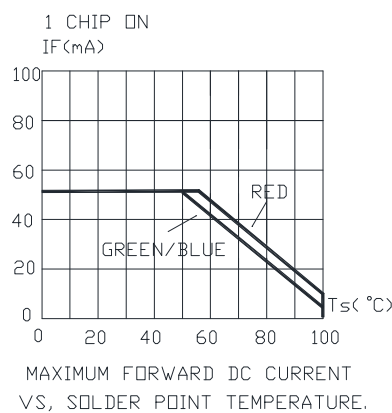
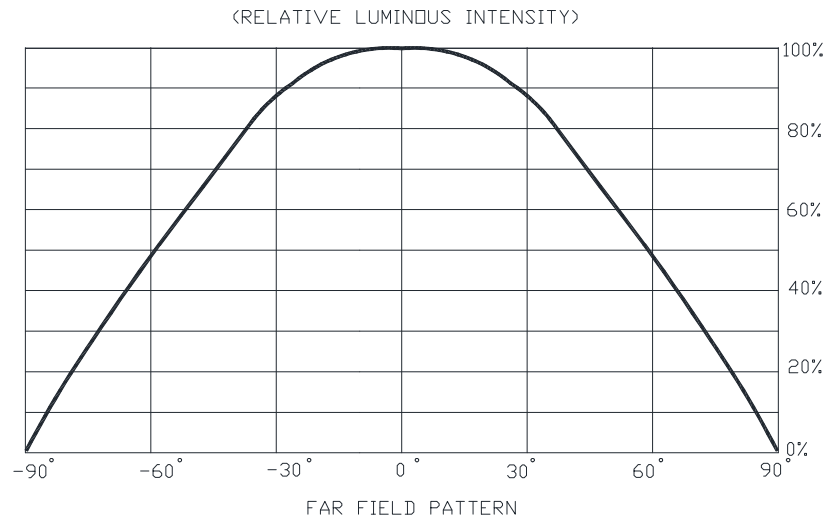
1. The above kit numbers represent the order codes which include multiple intensity-bin and color-bin codes. Only one intensity-bin code and one color-bin code will be shipped on each reel. Single intensity-bin code and single color bin code will be orderable in certain quantities. For example, any 1 intensity bin from K - N means only 1 intensity bin (K or M or N) will be shipped by Cree. For example, any 1 color bin from G7 - Ga means only 1 color bin (G7 or G8 or G9 or Ga) will be shipped by Cree.
2. Please refer to the "Cree LED Lamp Reliability Test Standards" document for reliability test conditions.
3. Please refer to the "Cree LED Lamp Soldering & Handling" document for information about how to use this LED product safely.

GRAPHS



The above data are collected from statistical figures that do not necessarily correspond to the actual parameters of each single LED. Hence, these data will be changed without further notice.

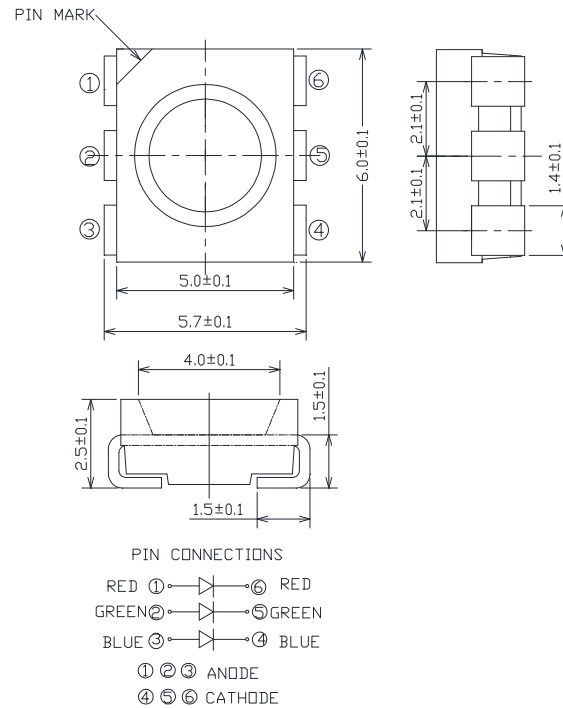
GRAPHS



The above data are collected from statistical figures that do not necessarily correspond to the actual parameters of each single LED. Hence, these data will be changed without further notice.

MECHANICAL DIMENSIONS

All dimensions are in mm.



NOTES

RoHS Compliance

The levels of environmentally sensitive, persistent biologically toxic (PBT), persistent organic pollutants (POP), or otherwise restricted materials in this product are below the maximum concentration values (also referred to as the threshold limits) permitted for such substances, or are used in an exempted application, in accordance with EU Directive 2002/95/EC on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS), as amended through April 21, 2006.

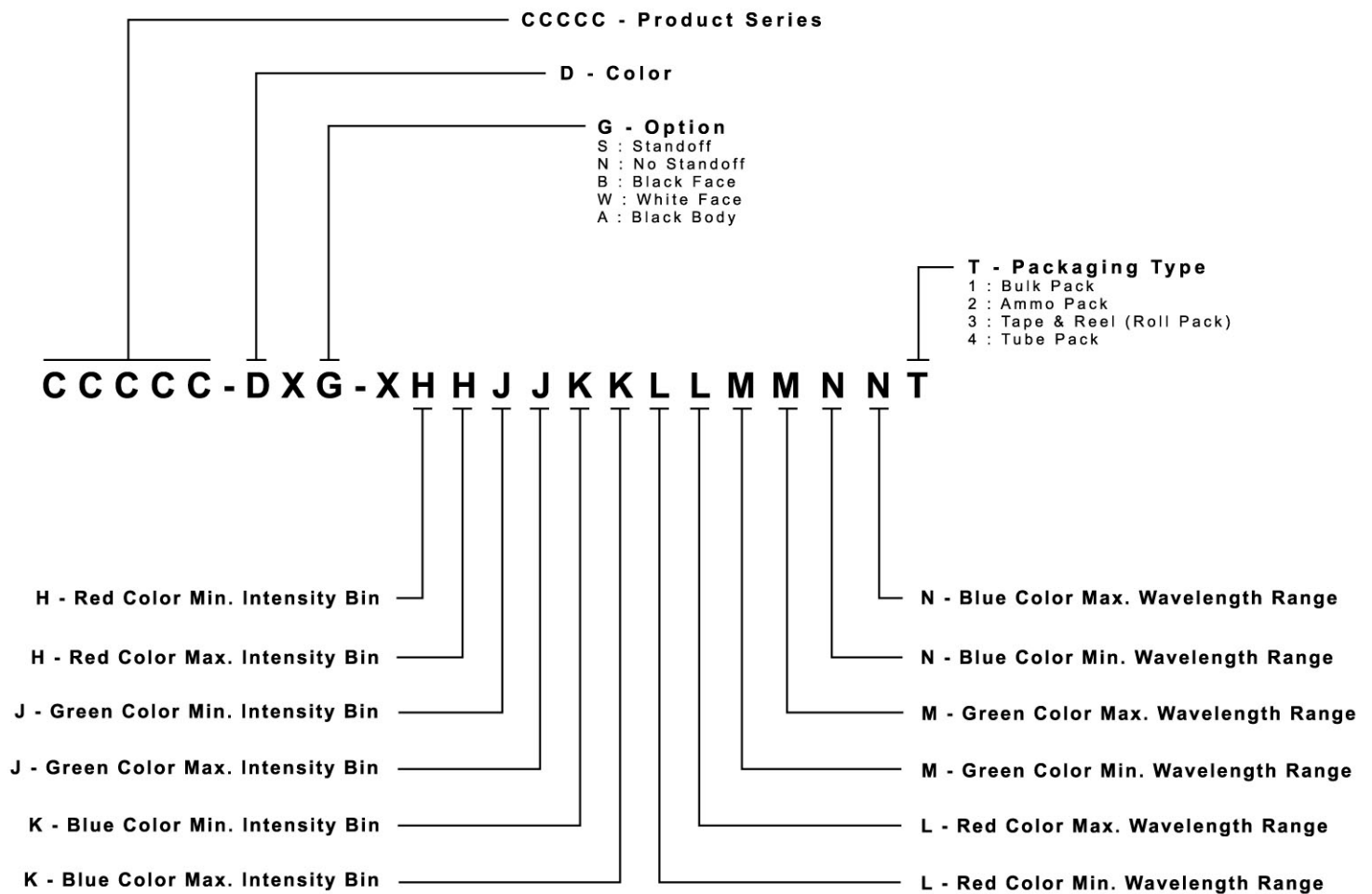
Vision Advisory Claim

Users should be cautioned not to stare at the light of this LED product. The bright light can damage the eye.

KIT NUMBER SYSTEM

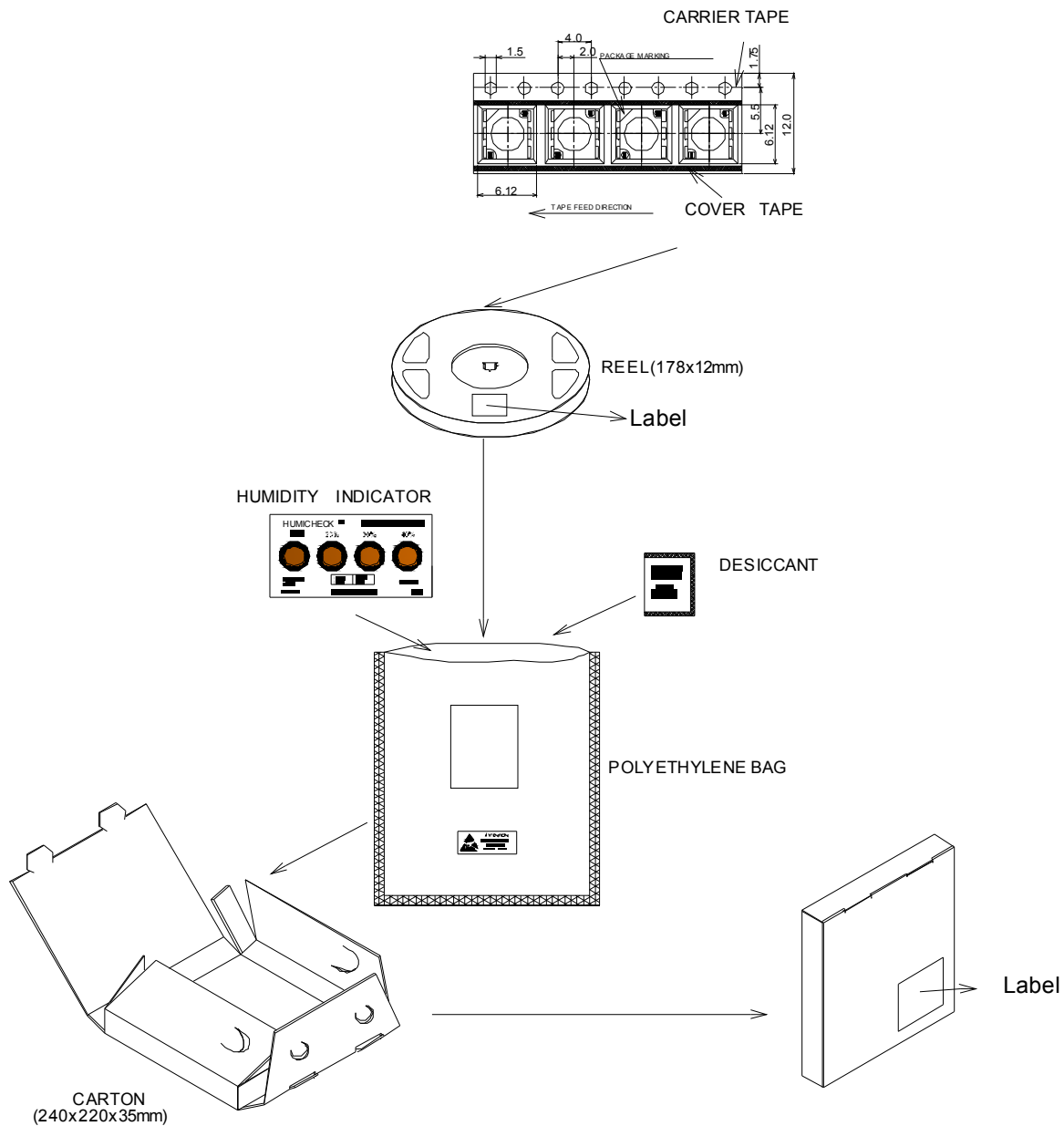
Cree LED lamps are tested and sorted into performance bins. A bin is specified by ranges of color, forward voltage, and brightness. Sorted LEDs are packaged for shipping in various convenient options. Please refer to the "Cree LED Lamp Packaging Standard" document for more information about shipping and packaging options.

Cree LEDs are sold by order codes in combinations of bins called kits. Order codes are configured in the following manner:



PACKAGING

- The boxes are not water resistant and they must be kept away from water and moisture.
- The LEDs are packed in cardboard boxes after packaging in normal or anti-electrostatic bags.
- Cardboard boxes will be used to protect the LEDs from mechanical shocks during transportation.
- The reel pack is applied in SMD LED.
- Max 900 pcs per reel.



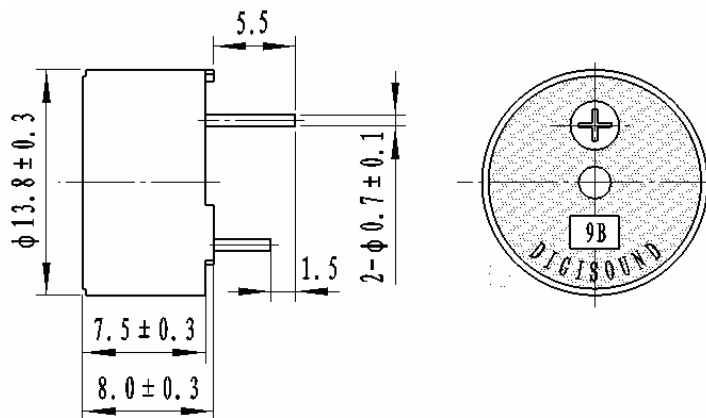
F/CM12P

Characteristics <i>Kennwerte</i>	Condition/Remark <i>Bedingungen/Kommentar</i>	Value <i>Wert</i>
Dimension <i>Abmessungen</i>	ø x H	13,8 x 7,5 mm
Rated voltage <i>Nennspannung</i>	DC	12 V
Operating voltage <i>Betriebsspannung</i>	DC	3...16 V
Rated current <i>Nennstrom</i>	Note 1	≤ 12 mA
Sound pressure level <i>Lautstärke</i>	Note 1	85 dB
Resonant frequency <i>Resonanzfrequenz</i>	+ 10%	4k±500 Hz
Operating temperature <i>Betriebstemperatur</i>		-20...+70°C
Storage temperature <i>Lagertemperatur</i>		-30...+80°C
Strength of Pin <i>Pin-Festigkeit</i>		1 kg
Weight <i>Gewicht</i>		0,8g
ROHS compliant		Yes
Housing material <i>Gehäusematerial</i>		PBT
Polarity marking	“+”-sign at the top of housing	
<i>Polaritätskennzeichnung</i>	„+“-Zeichen auf dem Gehäuse	

Note 1: Value applying rated voltage (DC)

DIMENSIONS (UNIT: mm)

Tolerance: ±0.5mm Except Specified



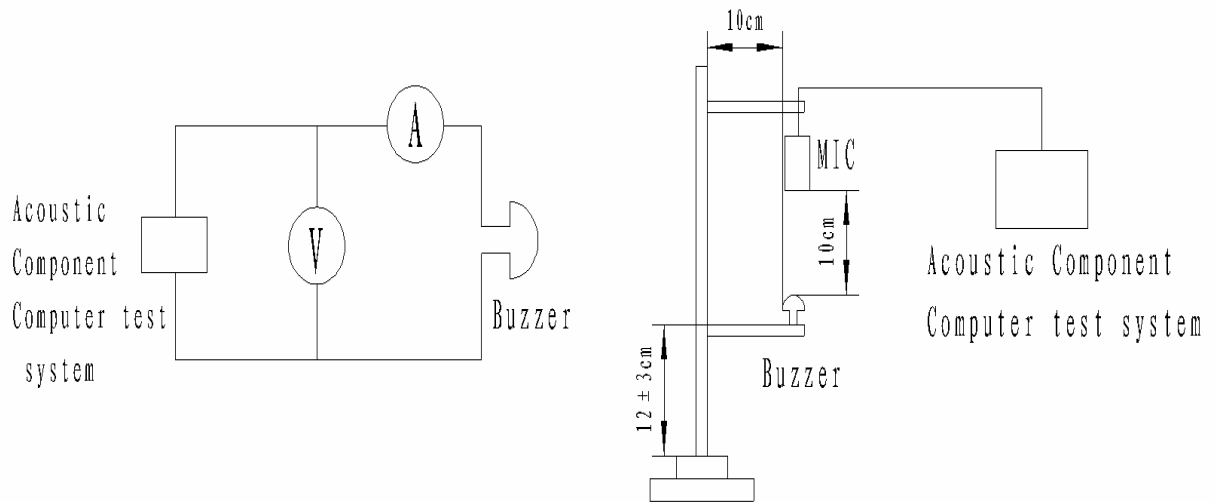
			Halbzeug, Werkstoff				Untolerierte Maße		Zeichn.-Nr. 090924.1H
							Maßstab		
	Tag	Name	And. zust.	And.-Mittlg. Nr.	Tag	Name	F/CM12P		
gezeichnet									
bearbeitet	24.09.09	KH							
geprüft									
normgepr.									

Diese Zeichnung ist unser Eigentum. Vervielfältigung unbefugte Verwertung, Mitteilung an andere ist strafbar und Schadensersatzpflichtig

Vervielfält.-Pause Nr.

Arbeitspause Nr.

TEST METHOD:

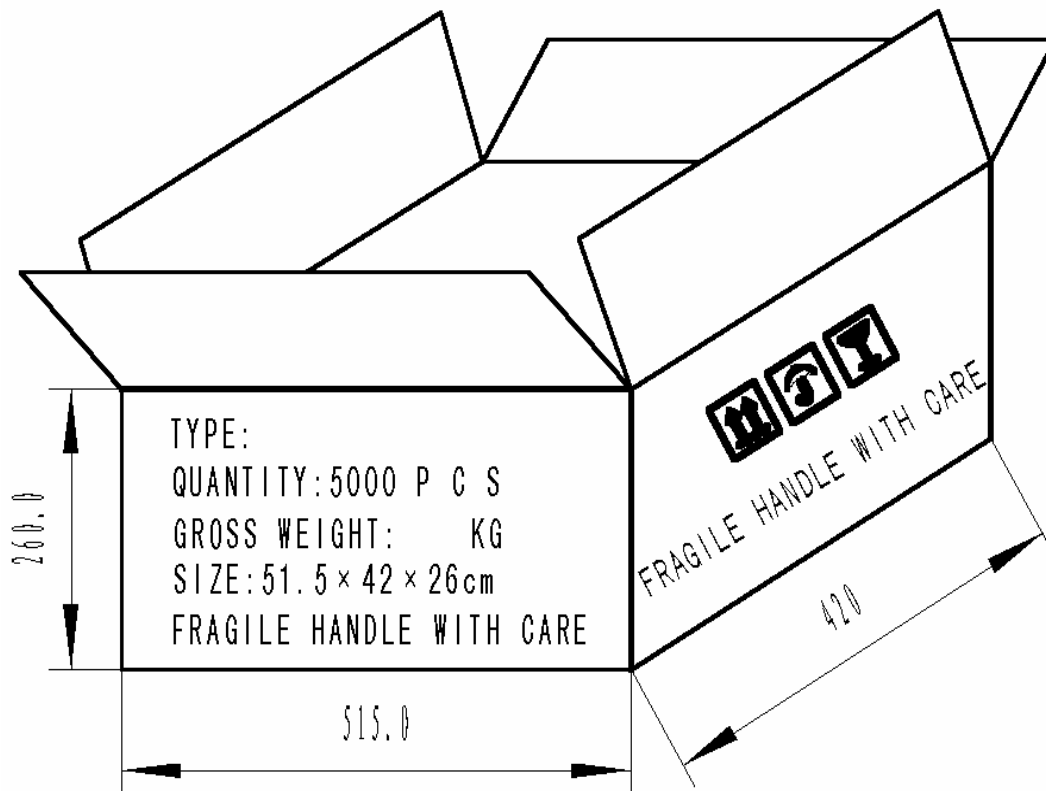


RELIABLY TEST:

NO.	ITEM	TESTING CONDITION	VARIANCE AFTER TEST
1	Humidity	50±5°C, 93(+2/-3)%RH, 96HRS	All specifications must be satisfied after the test.
2	<u>High temp.</u>	+80±2°C, 96HRS	
3	Low temp.	-30±2°C, 96HRS	
4	Temperature cycling	-30±2°C, 30minutes room temp. 15minutes +80±2°C, 30minutes room temp. 15minutes 5 cycles	
5	Drop test	3 times from height of 70cm onto the surface of 10mm thick wooden board.	
6	Vibration test	Make the test for the directions of X Y and Z (total 0.5 hours). To-and-fro. sweep time (from 10 to 55 Hz and then 55 to 10) under single amplitude of 10mm is 3minute.	
7	Solder heat resistance	The part leads (pins) shall be immersed in molten solder maintained at 250±10°C for a period of 30 seconds.	After the test part shall meet specifications without any degradation in appearance and performance.

		Halbzeug, Werkstoff		Untolerierte Maße		Zeichn.-Nr. 090924.1H				
				Maßstab						
Vervielfält.-Pause Nr.		Ersatz f. Zeichn.		F/CM12P						
Arbeitspause Nr.										
	Tag	Name	And. zust.					And.-Mittlg. Nr.	Tag	Name
gezeichnet										
bearbeitet	24.09.09	KH								
geprüft										
normgepr.										

PACKING:



NOTES:

1. 100 PCS per box
2. Total 50 boxes per carton
3. Total 5000 PCS carton
4. Volume: 51.5 x 42 x 26 cm

Diese Zeichnung ist unser Eigentum. Vervielfältigung unbefugte Verwertung, Mitteilung an andere ist strafbar und Schadensersatzpflichtig

		Halbzeug, Werkstoff				Untolerierte Maße		Zeichn.-Nr.	
						Maßstab		090924.1H	
Vervielfält.-Pause Nr.								Ersatz f. Zeichn.	
	Tag	Name	And. zust.	And.-Mittlg. Nr.	Tag	Name	F/CM12P		
gezeichnet									
bearbeitet	24.09.09	KH							
geprüft									
Arbeitspause Nr.									

FEATURES

- 3-axis sensing**
- Small, low profile package**
4 mm $\times$ 4 mm $\times$ 1.45 mm LFCSP
- Low power : 350 μ A (typical)**
- Single-supply operation: 1.8 V to 3.6 V**
- 10,000 g shock survival**
- Excellent temperature stability**
- BW adjustment with a single capacitor per axis**
- RoHS/WEEE lead-free compliant**

APPLICATIONS

- Cost sensitive, low power, motion- and tilt-sensing applications**
- Mobile devices**
- Gaming systems**
- Disk drive protection**
- Image stabilization**
- Sports and health devices**

GENERAL DESCRIPTION

The ADXL335 is a small, thin, low power, complete 3-axis accelerometer with signal conditioned voltage outputs. The product measures acceleration with a minimum full-scale range of $\pm 3 g$. It can measure the static acceleration of gravity in tilt-sensing applications, as well as dynamic acceleration resulting from motion, shock, or vibration.

The user selects the bandwidth of the accelerometer using the C_X , C_Y , and C_Z capacitors at the X_{OUT} , Y_{OUT} , and Z_{OUT} pins. Bandwidths can be selected to suit the application, with a range of 0.5 Hz to 1600 Hz for the X and Y axes, and a range of 0.5 Hz to 550 Hz for the Z axis.

The ADXL335 is available in a small, low profile, 4 mm $\times$ 4 mm $\times$ 1.45 mm, 16-lead, plastic lead frame chip scale package (LFCSP_LQ).

FUNCTIONAL BLOCK DIAGRAM

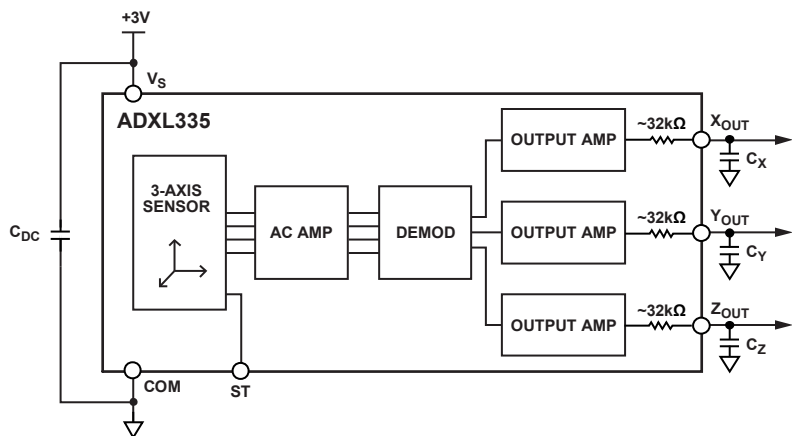


Figure 1.

Rev. B

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REVISION HISTORY

1/10—Rev. A to Rev. B

Changes to Figure 21	9
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7/09—Rev. 0 to Rev. A

Changes to Figure 22	9
Changes to Outline Dimensions	14

1/09—Revision 0: Initial Version

SPECIFICATIONS

$T_A = 25^\circ\text{C}$, $V_S = 3\text{ V}$, $C_X = C_Y = C_Z = 0.1\text{ }\mu\text{F}$, acceleration = 0 g, unless otherwise noted. All minimum and maximum specifications are guaranteed. Typical specifications are not guaranteed.

Table 1.

Parameter	Conditions	Min	Typ	Max	Unit
SENSOR INPUT	Each axis				
Measurement Range		± 3	± 3.6		g
Nonlinearity	% of full scale		± 0.3		%
Package Alignment Error			± 1		Degrees
Interaxis Alignment Error			± 0.1		Degrees
Cross-Axis Sensitivity ¹			± 1		%
SENSITIVITY (RATIOMETRIC) ²	Each axis				
Sensitivity at X_{OUT} , Y_{OUT} , Z_{OUT}	$V_S = 3\text{ V}$	270	300	330	mV/g
Sensitivity Change Due to Temperature ³	$V_S = 3\text{ V}$		± 0.01		%/ $^\circ\text{C}$
ZERO g BIAS LEVEL (RATIOMETRIC)					
0 g Voltage at X_{OUT} , Y_{OUT}	$V_S = 3\text{ V}$	1.35	1.5	1.65	V
0 g Voltage at Z_{OUT}	$V_S = 3\text{ V}$	1.2	1.5	1.8	V
0 g Offset vs. Temperature			± 1		mg/ $^\circ\text{C}$
NOISE PERFORMANCE					
Noise Density X_{OUT} , Y_{OUT}			150		$\mu\text{g}/\sqrt{\text{Hz}}$ rms
Noise Density Z_{OUT}			300		$\mu\text{g}/\sqrt{\text{Hz}}$ rms
FREQUENCY RESPONSE ⁴					
Bandwidth X_{OUT} , Y_{OUT} ⁵	No external filter		1600		Hz
Bandwidth Z_{OUT} ⁵	No external filter		550		Hz
R_{FILT} Tolerance			$32 \pm 15\%$		k Ω
Sensor Resonant Frequency			5.5		kHz
SELF-TEST ⁶					
Logic Input Low			+0.6		V
Logic Input High			+2.4		V
ST Actuation Current			+60		μA
Output Change at X_{OUT}	Self-Test 0 to Self-Test 1	-150	-325	-600	mV
Output Change at Y_{OUT}	Self-Test 0 to Self-Test 1	+150	+325	+600	mV
Output Change at Z_{OUT}	Self-Test 0 to Self-Test 1	+150	+550	+1000	mV
OUTPUT AMPLIFIER					
Output Swing Low	No load		0.1		V
Output Swing High	No load		2.8		V
POWER SUPPLY					
Operating Voltage Range		1.8		3.6	V
Supply Current	$V_S = 3\text{ V}$		350		μA
Turn-On Time ⁷	No external filter		1		ms
TEMPERATURE					
Operating Temperature Range		-40		+85	$^\circ\text{C}$

¹ Defined as coupling between any two axes.

² Sensitivity is essentially ratiometric to V_S .

³ Defined as the output change from ambient-to-maximum temperature or ambient-to-minimum temperature.

⁴ Actual frequency response controlled by user-supplied external filter capacitors (C_X , C_Y , C_Z).

⁵ Bandwidth with external capacitors = $1/(2 \times \pi \times 32\text{ k}\Omega \times C)$. For C_X , $C_Y = 0.003\text{ }\mu\text{F}$, bandwidth = 1.6 kHz. For $C_Z = 0.01\text{ }\mu\text{F}$, bandwidth = 500 Hz. For C_X , C_Y , $C_Z = 10\text{ }\mu\text{F}$, bandwidth = 0.5 Hz.

⁶ Self-test response changes cubically with V_S .

⁷ Turn-on time is dependent on C_X , C_Y , C_Z and is approximately $160 \times C_X$ or C_Y or $C_Z + 1\text{ ms}$, where C_X , C_Y , C_Z are in microfarads (μF).

ABSOLUTE MAXIMUM RATINGS

Table 2.

Parameter	Rating
Acceleration (Any Axis, Unpowered)	10,000 <i>g</i>
Acceleration (Any Axis, Powered)	10,000 <i>g</i>
V _s	–0.3 V to +3.6 V
All Other Pins	(COM – 0.3 V) to (V _s + 0.3 V)
Output Short-Circuit Duration (Any Pin to Common)	Indefinite
Temperature Range (Powered)	–55°C to +125°C
Temperature Range (Storage)	–65°C to +150°C

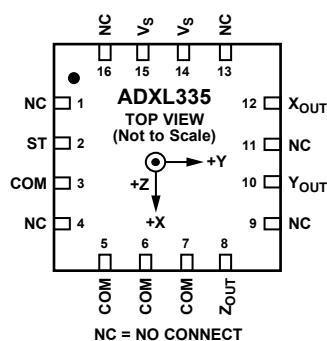
Stresses above those listed under Absolute Maximum Ratings may cause permanent damage to the device. This is a stress rating only; functional operation of the device at these or any other conditions above those indicated in the operational section of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

ESD CAUTION



ESD (electrostatic discharge) sensitive device. Charged devices and circuit boards can discharge without detection. Although this product features patented or proprietary protection circuitry, damage may occur on devices subjected to high energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality.

PIN CONFIGURATION AND FUNCTION DESCRIPTIONS



NOTES
1. EXPOSED PAD IS NOT INTERNALLY CONNECTED BUT SHOULD BE SOLDERED FOR MECHANICAL INTEGRITY.

077008-003

Figure 2. Pin Configuration

Table 3. Pin Function Descriptions

Pin No.	Mnemonic	Description
1	NC	No Connect. ¹
2	ST	Self-Test.
3	COM	Common.
4	NC	No Connect. ¹
5	COM	Common.
6	COM	Common.
7	COM	Common.
8	Z _{OUT}	Z Channel Output.
9	NC	No Connect. ¹
10	Y _{OUT}	Y Channel Output.
11	NC	No Connect. ¹
12	X _{OUT}	X Channel Output.
13	NC	No Connect. ¹
14	V _S	Supply Voltage (1.8 V to 3.6 V).
15	V _S	Supply Voltage (1.8 V to 3.6 V).
16	NC	No Connect. ¹
EP	Exposed Pad	Not internally connected. Solder for mechanical integrity.

¹ NC pins are not internally connected and can be tied to COM pins, unless otherwise noted.

TYPICAL PERFORMANCE CHARACTERISTICS

N > 1000 for all typical performance plots, unless otherwise noted.

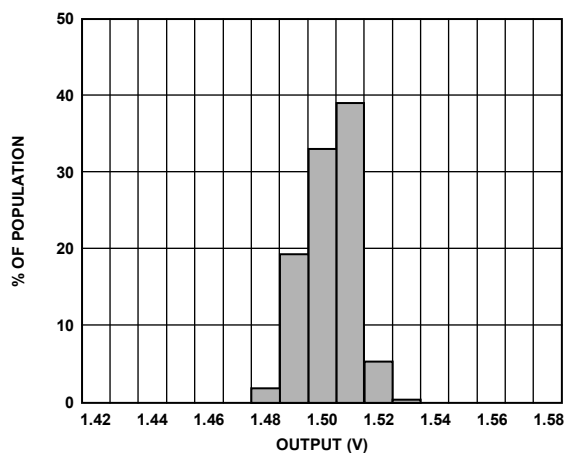


Figure 3. X-Axis Zero g Bias at 25°C, $V_S = 3\text{ V}$

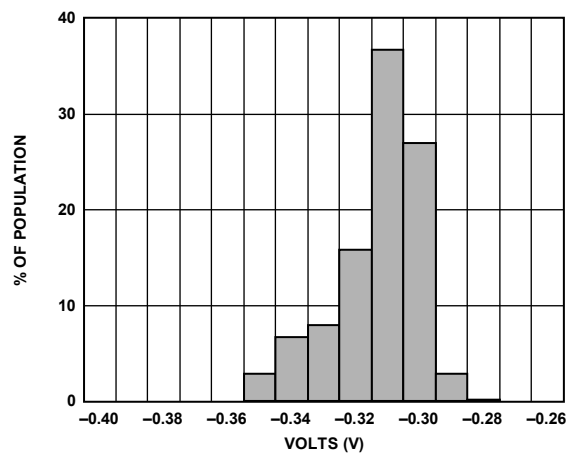


Figure 6. X-Axis Self-Test Response at 25°C, $V_S = 3\text{ V}$

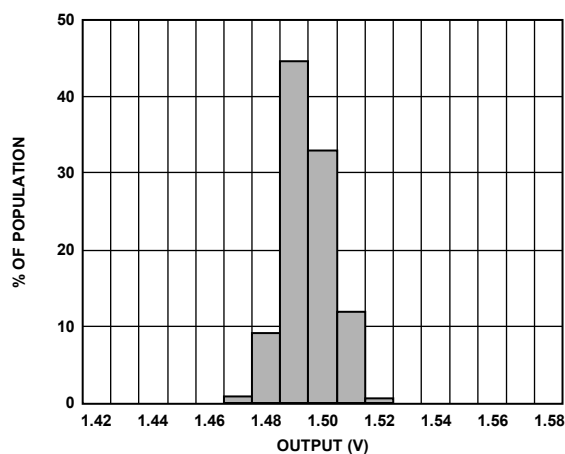


Figure 4. Y-Axis Zero g Bias at 25°C, $V_S = 3\text{ V}$

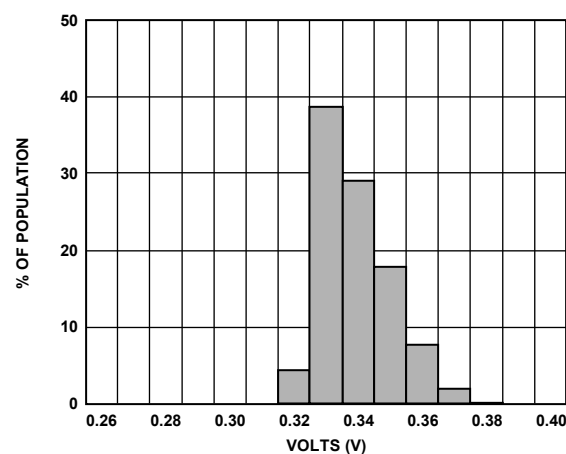


Figure 7. Y-Axis Self-Test Response at 25°C, $V_S = 3\text{ V}$

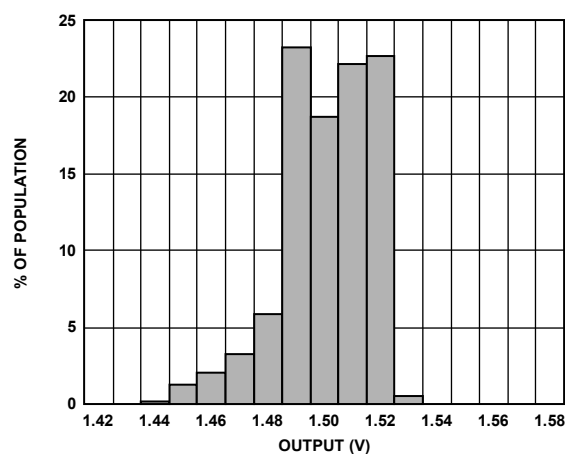


Figure 5. Z-Axis Zero g Bias at 25°C, $V_S = 3\text{ V}$

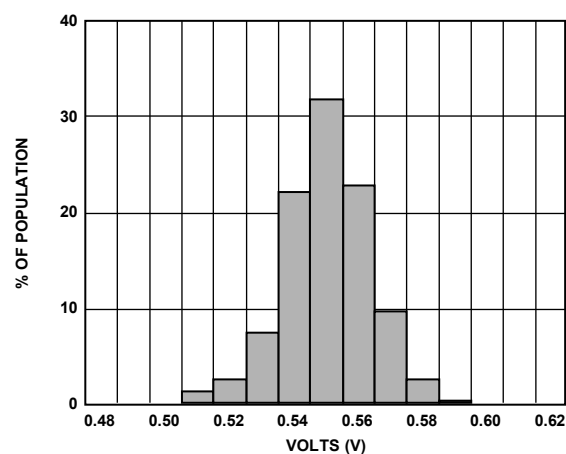


Figure 8. Z-Axis Self-Test Response at 25°C, $V_S = 3\text{ V}$

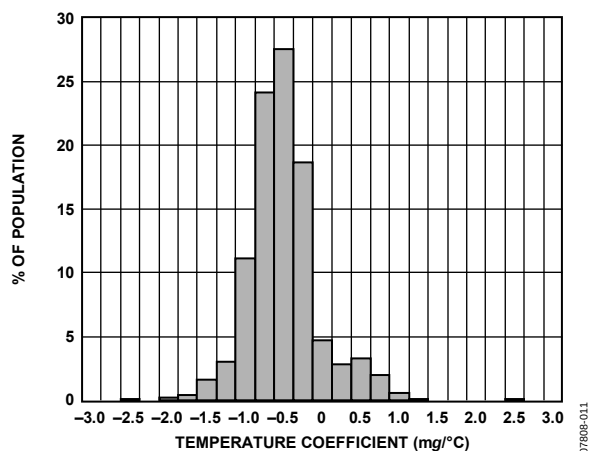


Figure 9. X-Axis Zero g Bias Temperature Coefficient, $V_S = 3\text{ V}$

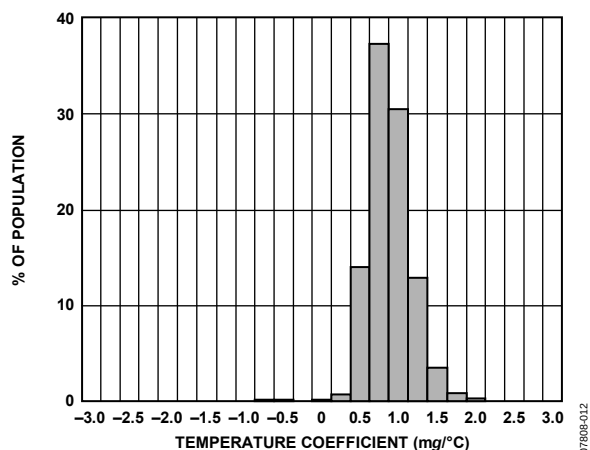


Figure 10. Y-Axis Zero g Bias Temperature Coefficient, $V_S = 3\text{ V}$

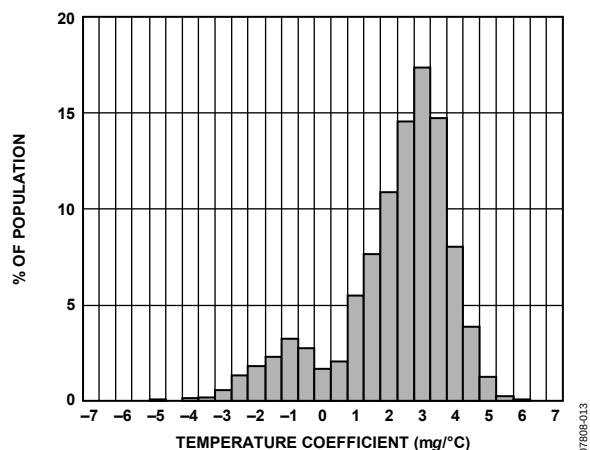


Figure 11. Z-Axis Zero g Bias Temperature Coefficient, $V_S = 3\text{ V}$

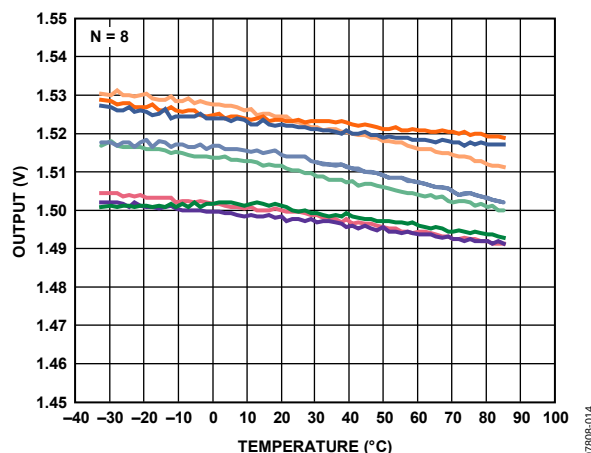


Figure 12. X-Axis Zero g Bias vs. Temperature—
Eight Parts Soldered to PCB

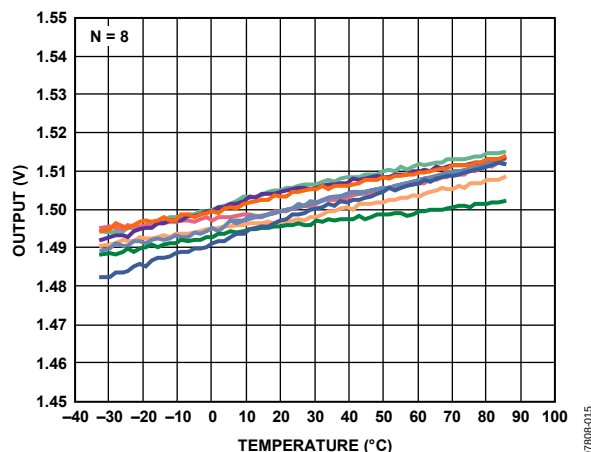


Figure 13. Y-Axis Zero g Bias vs. Temperature—
Eight Parts Soldered to PCB

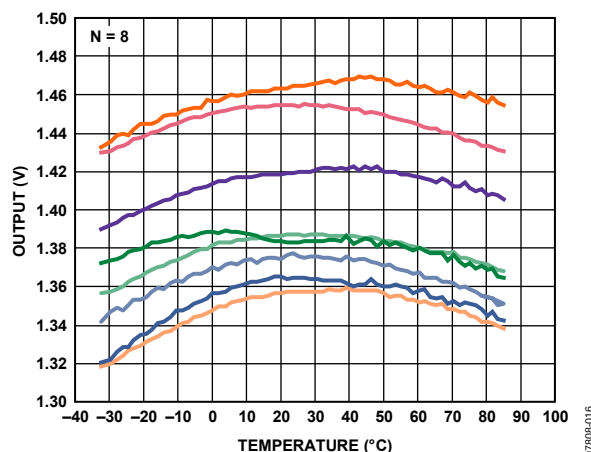


Figure 14. Z-Axis Zero g Bias vs. Temperature—
Eight Parts Soldered to PCB

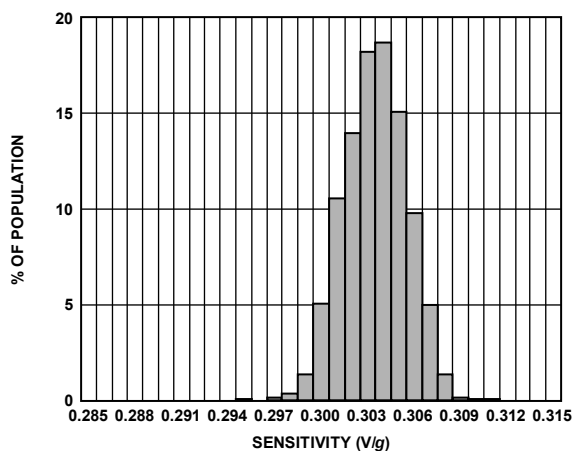


Figure 15. X-Axis Sensitivity at 25°C, $V_S = 3\text{ V}$

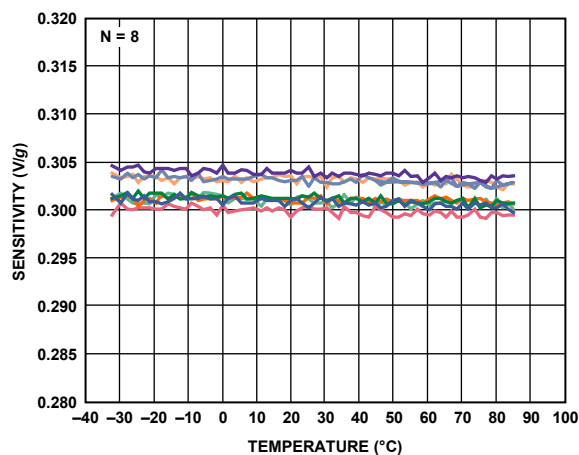


Figure 18. X-Axis Sensitivity vs. Temperature—
Eight Parts Soldered to PCB, $V_S = 3\text{ V}$

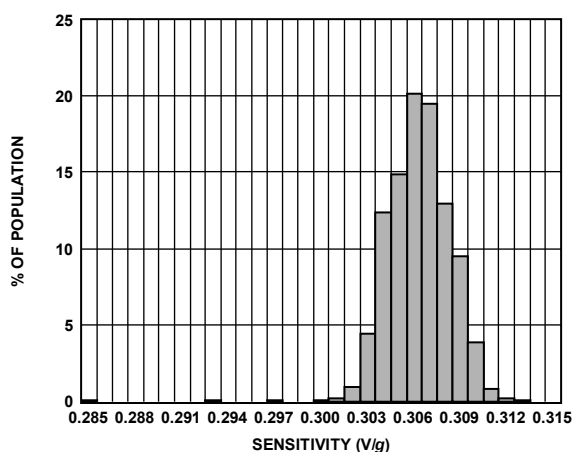


Figure 16. Y-Axis Sensitivity at 25°C, $V_S = 3\text{ V}$

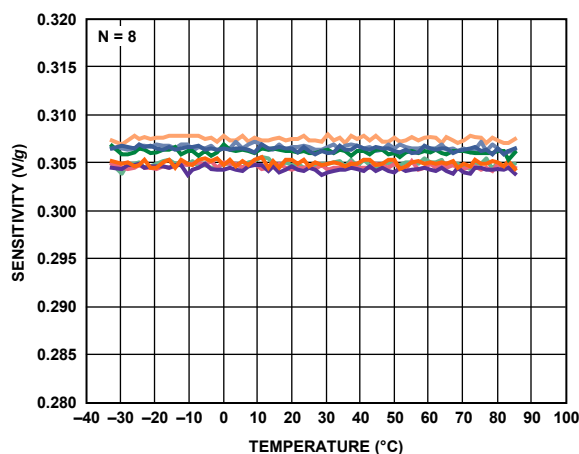


Figure 19. Y-Axis Sensitivity vs. Temperature—
Eight Parts Soldered to PCB, $V_S = 3\text{ V}$

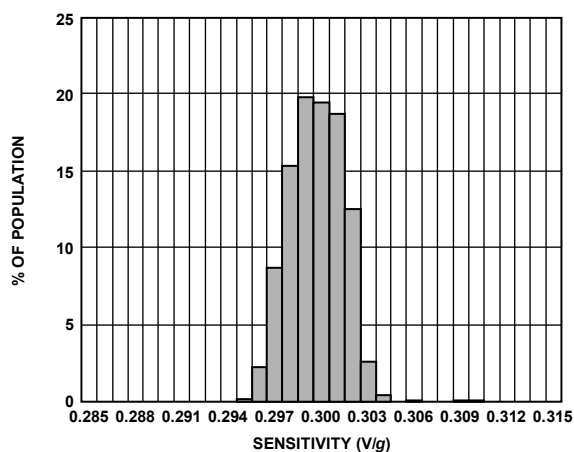


Figure 17. Z-Axis Sensitivity at 25°C, $V_S = 3\text{ V}$

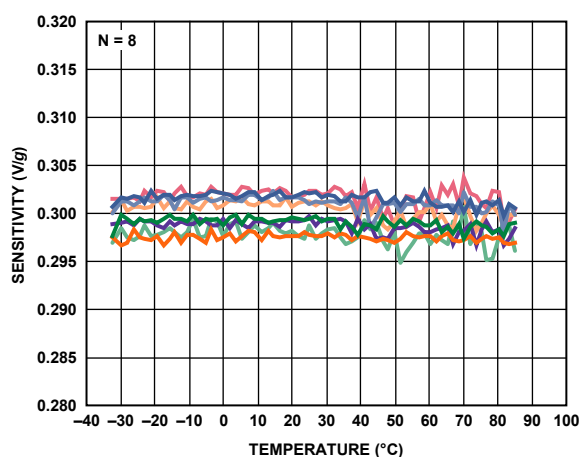


Figure 20. Z-Axis Sensitivity vs. Temperature—
Eight Parts Soldered to PCB, $V_S = 3\text{ V}$

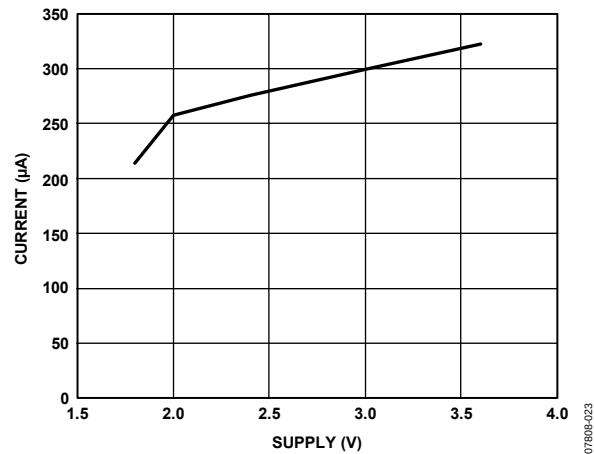


Figure 21. Typical Current Consumption vs. Supply Voltage

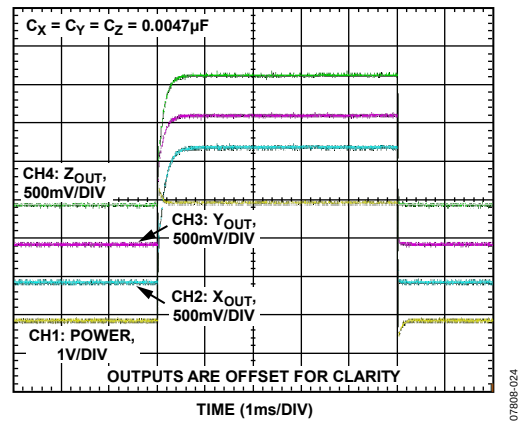


Figure 22. Typical Turn-On Time, $V_S = 3V$

THEORY OF OPERATION

The ADXL335 is a complete 3-axis acceleration measurement system. The ADXL335 has a measurement range of ± 3 g minimum. It contains a polysilicon surface-micromachined sensor and signal conditioning circuitry to implement an open-loop acceleration measurement architecture. The output signals are analog voltages that are proportional to acceleration. The accelerometer can measure the static acceleration of gravity in tilt-sensing applications as well as dynamic acceleration resulting from motion, shock, or vibration.

The sensor is a polysilicon surface-micromachined structure built on top of a silicon wafer. Polysilicon springs suspend the structure over the surface of the wafer and provide a resistance against acceleration forces. Deflection of the structure is measured using a differential capacitor that consists of independent fixed plates and plates attached to the moving mass. The fixed plates are driven by 180° out-of-phase square waves. Acceleration deflects the moving mass and unbalances the differential capacitor resulting in a sensor output whose amplitude is proportional to acceleration. Phase-sensitive demodulation techniques are then used to determine the magnitude and direction of the acceleration.

The demodulator output is amplified and brought off-chip through a $32\text{ k}\Omega$ resistor. The user then sets the signal bandwidth of the device by adding a capacitor. This filtering improves measurement resolution and helps prevent aliasing.

MECHANICAL SENSOR

The ADXL335 uses a single structure for sensing the X, Y, and Z axes. As a result, the three axes' sense directions are highly orthogonal and have little cross-axis sensitivity. Mechanical misalignment of the sensor die to the package is the chief source of cross-axis sensitivity. Mechanical misalignment can, of course, be calibrated out at the system level.

PERFORMANCE

Rather than using additional temperature compensation circuitry, innovative design techniques ensure that high performance is built in to the ADXL335. As a result, there is no quantization error or nonmonotonic behavior, and temperature hysteresis is very low (typically less than 3 mg over the -25°C to $+70^\circ\text{C}$ temperature range).

APPLICATIONS INFORMATION

POWER SUPPLY DECOUPLING

For most applications, a single 0.1 μF capacitor, C_{DC} , placed close to the ADXL335 supply pins adequately decouples the accelerometer from noise on the power supply. However, in applications where noise is present at the 50 kHz internal clock frequency (or any harmonic thereof), additional care in power supply bypassing is required because this noise can cause errors in acceleration measurement.

If additional decoupling is needed, a 100 Ω (or smaller) resistor or ferrite bead can be inserted in the supply line. Additionally, a larger bulk bypass capacitor (1 μF or greater) can be added in parallel to C_{DC} . Ensure that the connection from the ADXL335 ground to the power supply ground is low impedance because noise transmitted through ground has a similar effect to noise transmitted through V_{S} .

SETTING THE BANDWIDTH USING C_{X} , C_{Y} , AND C_{Z}

The ADXL335 has provisions for band limiting the X_{OUT} , Y_{OUT} , and Z_{OUT} pins. Capacitors must be added at these pins to implement low-pass filtering for antialiasing and noise reduction. The equation for the 3 dB bandwidth is

$$F_{-3\text{ dB}} = 1/(2\pi(32\text{ k}\Omega) \times C_{(\text{X}, \text{Y}, \text{Z})})$$

or more simply

$$F_{-3\text{ dB}} = 5\text{ }\mu\text{F}/C_{(\text{X}, \text{Y}, \text{Z})}$$

The tolerance of the internal resistor (R_{FILT}) typically varies as much as $\pm 15\%$ of its nominal value (32 k Ω), and the bandwidth varies accordingly. A minimum capacitance of 0.0047 μF for C_{X} , C_{Y} , and C_{Z} is recommended in all cases.

Table 4. Filter Capacitor Selection, C_{X} , C_{Y} , and C_{Z}

Bandwidth (Hz)	Capacitor (μF)
1	4.7
10	0.47
50	0.10
100	0.05
200	0.027
500	0.01

SELF-TEST

The ST pin controls the self-test feature. When this pin is set to V_{S} , an electrostatic force is exerted on the accelerometer beam. The resulting movement of the beam allows the user to test if the accelerometer is functional. The typical change in output is -1.08 g (corresponding to -325 mV) in the X-axis, $+1.08\text{ g}$ (or $+325\text{ mV}$) on the Y-axis, and $+1.83\text{ g}$ (or $+550\text{ mV}$) on the Z-axis. This ST pin can be left open-circuit or connected to common (COM) in normal use.

Never expose the ST pin to voltages greater than $V_{\text{S}} + 0.3\text{ V}$. If this cannot be guaranteed due to the system design (for instance, if there are multiple supply voltages), then a low V_{F} clamping diode between ST and V_{S} is recommended.

DESIGN TRADE-OFFS FOR SELECTING FILTER CHARACTERISTICS: THE NOISE/BW TRADE-OFF

The selected accelerometer bandwidth ultimately determines the measurement resolution (smallest detectable acceleration). Filtering can be used to lower the noise floor to improve the resolution of the accelerometer. Resolution is dependent on the analog filter bandwidth at X_{OUT} , Y_{OUT} , and Z_{OUT} .

The output of the ADXL335 has a typical bandwidth of greater than 500 Hz. The user must filter the signal at this point to limit aliasing errors. The analog bandwidth must be no more than half the analog-to-digital sampling frequency to minimize aliasing. The analog bandwidth can be further decreased to reduce noise and improve resolution.

The ADXL335 noise has the characteristics of white Gaussian noise, which contributes equally at all frequencies and is described in terms of $\mu\text{g}/\sqrt{\text{Hz}}$ (the noise is proportional to the square root of the accelerometer bandwidth). The user should limit bandwidth to the lowest frequency needed by the application to maximize the resolution and dynamic range of the accelerometer.

With the single-pole, roll-off characteristic, the typical noise of the ADXL335 is determined by

$$\text{rms Noise} = \text{Noise Density} \times (\sqrt{\text{BW} \times 1.6})$$

It is often useful to know the peak value of the noise. Peak-to-peak noise can only be estimated by statistical methods. Table 5 is useful for estimating the probabilities of exceeding various peak values, given the rms value.

Table 5. Estimation of Peak-to-Peak Noise

Peak-to-Peak Value	% of Time That Noise Exceeds Nominal Peak-to-Peak Value
$2 \times \text{rms}$	32
$4 \times \text{rms}$	4.6
$6 \times \text{rms}$	0.27
$8 \times \text{rms}$	0.006

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USE WITH OPERATING VOLTAGES OTHER THAN 3 V

The ADXL335 is tested and specified at $V_S = 3\text{ V}$; however, it can be powered with V_S as low as 1.8 V or as high as 3.6 V . Note that some performance parameters change as the supply voltage is varied.

The ADXL335 output is ratiometric, therefore, the output sensitivity (or scale factor) varies proportionally to the supply voltage. At $V_S = 3.6\text{ V}$, the output sensitivity is typically 360 mV/g . At $V_S = 2\text{ V}$, the output sensitivity is typically 195 mV/g .

The zero g bias output is also ratiometric, thus the zero g output is nominally equal to $V_S/2$ at all supply voltages.

The output noise is not ratiometric but is absolute in volts; therefore, the noise density decreases as the supply voltage increases. This is because the scale factor (mV/g) increases while the noise voltage remains constant. At $V_S = 3.6\text{ V}$, the X-axis and Y-axis noise density is typically $120\text{ }\mu\text{g}/\sqrt{\text{Hz}}$, whereas at $V_S = 2\text{ V}$, the X-axis and Y-axis noise density is typically $270\text{ }\mu\text{g}/\sqrt{\text{Hz}}$.

Self-test response in g is roughly proportional to the square of the supply voltage. However, when ratiometricity of sensitivity is factored in with supply voltage, the self-test response in volts is roughly proportional to the cube of the supply voltage. For example, at $V_S = 3.6\text{ V}$, the self-test response for the ADXL335 is approximately -560 mV for the X-axis, $+560\text{ mV}$ for the Y-axis, and $+950\text{ mV}$ for the Z-axis.

At $V_S = 2\text{ V}$, the self-test response is approximately -96 mV for the X-axis, $+96\text{ mV}$ for the Y-axis, and -163 mV for the Z-axis.

The supply current decreases as the supply voltage decreases. Typical current consumption at $V_S = 3.6\text{ V}$ is $375\text{ }\mu\text{A}$, and typical current consumption at $V_S = 2\text{ V}$ is $200\text{ }\mu\text{A}$.

AXES OF ACCELERATION SENSITIVITY

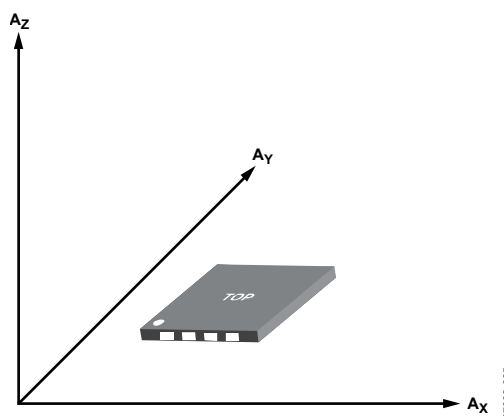


Figure 23. Axes of Acceleration Sensitivity; Corresponding Output Voltage Increases When Accelerated Along the Sensitive Axis.

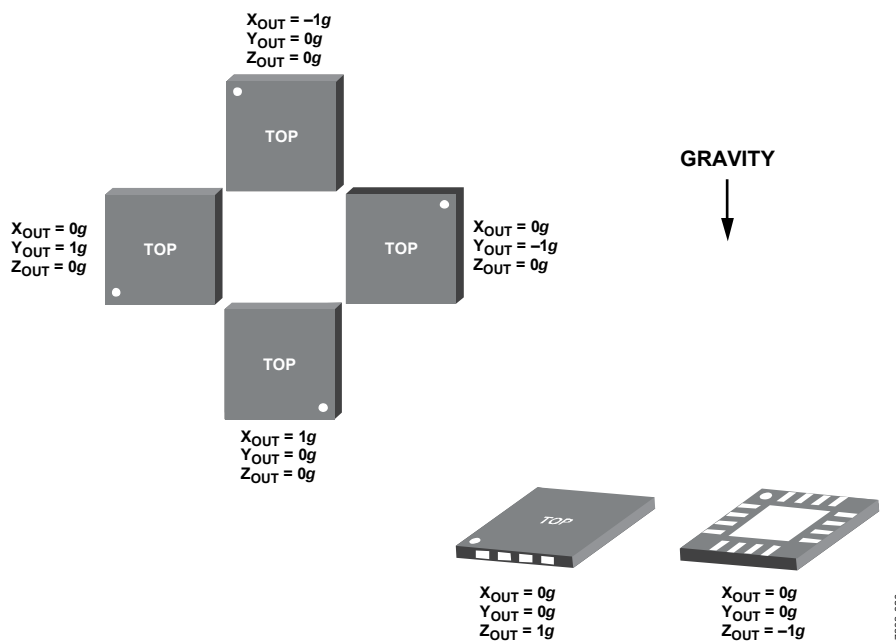


Figure 24. Output Response vs. Orientation to Gravity

LAYOUT AND DESIGN RECOMMENDATIONS

The recommended soldering profile is shown in Figure 25 followed by a description of the profile features in Table 6. The recommended PCB layout or solder land drawing is shown in Figure 26.

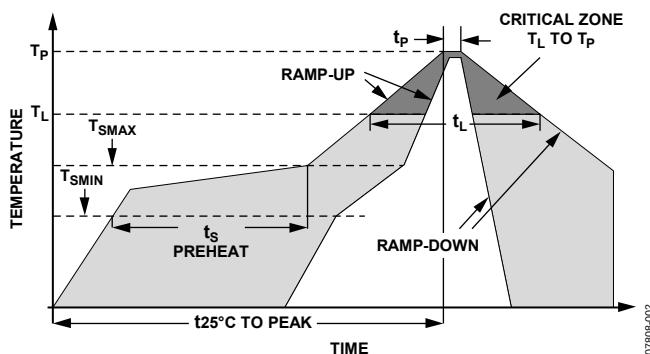


Figure 25. Recommended Soldering Profile

Table 6. Recommended Soldering Profile

Profile Feature	Sn63/Pb37	Pb-Free
Average Ramp Rate (T_L to T_P)	3°C/sec max	3°C/sec max
Preheat		
Minimum Temperature (T_{SMIN})	100°C	150°C
Maximum Temperature (T_{SMAX})	150°C	200°C
Time (T_{SMIN} to T_{SMAX}) (t_S)	60 sec to 120 sec	60 sec to 180 sec
T_{SMAX} to T_L		
Ramp-Up Rate	3°C/sec max	3°C/sec max
Time Maintained Above Liquidous (T_L)		
Liquidous Temperature (T_L)	183°C	217°C
Time (t_L)	60 sec to 150 sec	60 sec to 150 sec
Peak Temperature (T_P)	240°C + 0°C/-5°C	260°C + 0°C/-5°C
Time Within 5°C of Actual Peak Temperature (t_P)	10 sec to 30 sec	20 sec to 40 sec
Ramp-Down Rate	6°C/sec max	6°C/sec max
Time 25°C to Peak Temperature	6 minutes max	8 minutes max

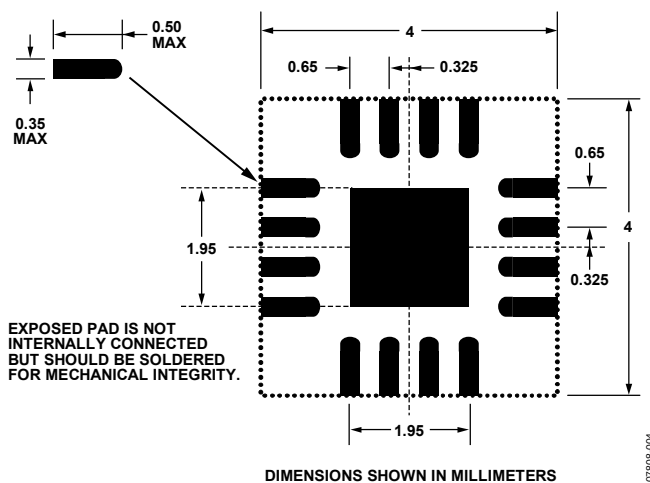
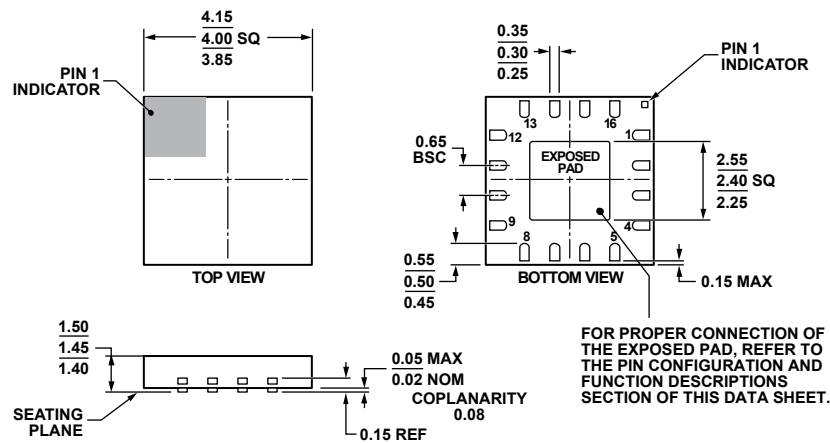


Figure 26. Recommended PCB Layout

OUTLINE DIMENSIONS



051902-A

ORDERING GUIDE

Model ¹	Measurement Range	Specified Voltage	Temperature Range	Package Description	Package Option
ADXL335BCPZ	±3 g	3 V	−40°C to +85°C	16-Lead LFCSP_LQ	CP-16-14
ADXL335BCPZ-RL	±3 g	3 V	−40°C to +85°C	16-Lead LFCSP_LQ	CP-16-14
ADXL335BCPZ-RL7	±3 g	3 V	−40°C to +85°C	16-Lead LFCSP_LQ	CP-16-14
EVAL-ADXL335Z				Evaluation Board	

¹ Z = RoHS Compliant Part.

NOTES

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NOTES

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